

1

2 **Supporting Information for**

3 **Informal Connections Outweigh Co-authorship Ties in Academic Impact**

4 **Lluís Danús, William Dinneen, Carolina Torreblanca, Guy Grossman and Sandra González-Bailón**

5 **Corresponding Author:**

6 **sgonzalezbailon@asc.upenn.edu**

7 **This PDF file includes:**

8 Figs. S1 to S42

9 Tables S1 to S18

10 SI References

1. List of Journals in our Sample

We obtained the full list of Political Science Journals from the Clarivate’s Social Sciences Index. We excluded non-English journals, journals with an impact factor lower than 1, and journals that do not use peer review. In Table S1 we print the full list of journals included in our sample ($N = 174$). In Table S5 we list an expanded version showing the number of papers analyzed for each journal, the fraction containing an acknowledgment section, and measures of journal prestige according to the SCImago (SJR) and CiteScore (CS) measures. We used the database Elsevier Scopus to obtain the index of all articles published in each of these journals, which we then downloaded as pdfs.

2. Data Extraction from Acknowledgment Sections

We created a pipeline to extract data from the acknowledgment sections if present in the pdfs of the articles. This pipeline includes three distinct steps: 1) locating the acknowledgment sections within the full-text articles; 2) extracting the names of scholars mentioned in the located section; and 3) matching those names with the author profiles in Scopus (when possible). A corpus of $N = 4,196$ hand-annotated acknowledgment sections served as the training set.

Acknowledgment Section Identification. (1) We stripped each article of numerals, punctuation, and lines shorter than five words or dominated by non-alphabetic characters; and (2) we segmented them into overlapping 1,000-character windows with a 500-character stride. We labeled ‘positive’ (acknowledgment) exactly one window per article, and labeled all others ‘negative’. To mitigate class imbalance, we down-sampled negative examples to a 10:1 ratio (24,380 negative and 2,438 positive). We represented windows as counts of the 500 most frequent 1–3-gram terms drawn from the acknowledgment text. We then trained a linear support-vector machine (SVM) with probability calibration under five-fold cross-validation ($F1 = 0.92$). We used this model to infer the index of acknowledgment sections in the entire corpus of papers.

Commenter Name Extraction. We supplied each candidate acknowledgment section to the gpt-4o-mini language model via OpenAI’s completion API alongside a structured, few-shot prompt (see Fig. S1) instructing the model to return a JSON record with seven fields: ‘commenters’, ‘reviewers’, ‘researchAssistants’, ‘organizations’, ‘seminars_or_workshops’, ‘fundingSources’, ‘otherPeople’, and a binary ‘includesAckSection’ flag to corroborate the SVM model used in step 1 of the pipeline. We requested three independent completions for every passage, and we retained the modal for each field. We discarded passages flagged as non-acknowledgment. When tested against the training set, the prompt extracted all commenter names in 96% of papers and introduced no spurious names in 78% of those cases. We excluded research assistants from our network data because these names are very difficult to disambiguate (see the following section) but, most importantly, because they are not as indicative of peer communication as of the amount of funding and monetary resources available to the authors through grants or funding support from their institutions. In Table S4 we show that 7.7% of the articles we analyze acknowledged at least one research assistant. Slightly more than half of these are classified as male (according to genderize.io).

Name Disambiguation. We normalized commenter names (i.e., we removed punctuation and performed common regular expression operations, like camel-case splitting, single-letter-surname correction, nickname expansion, and removal of diacritics) and we assigned them a persistent UUID prior to any disambiguation. We then queried each name against the Scopus Author Search API, initially restricting the search to the Social Sciences subject area. We classified search outcomes as unique (one profile), absent (zero profiles) or ambiguous (>1 profile). We accepted unique hits (48.2% of names) immediately; we marked absent names (28.1 %) first re-searched without the Social Sciences filter, and labeled them “not-on-Scopus” if they still yielded no results after removing the filter; for names in the ambiguous group (23.7 %), we ranked candidate profiles by the number of keyword overlaps with the focal article and we retained the top-ranked profile. This keyword-based filter raised the share of uniquely identified commenters to 60.8 %. We retained all commenters as nodes in the acknowledgment network, regardless of whether a Scopus profile could be assigned. The presence of a Scopus identifier allowed us to more confidently link commenters across multiple papers.

3. How Many Authors Acknowledge?

In the main text we note that more than half (55.61%) of the articles have an acknowledgment section. In Fig. S2 we show that authors who never acknowledge in their papers have a similar composition in terms of their gender, seniority level, geographical region, and methodological orientation: this subset of authors is predominantly male, senior, based in Europe, and of a quantitative orientation (note that the histograms in each panel are overlaid, to simplify visualization). In total, authors who acknowledge at least once are a majority.

In Fig. S3 we focus on paper-level counts. Papers with an acknowledgment section tend to have a quantitative orientation (see also section 6) and they are written by multiple authors. The largest proportion of articles with no acknowledgment section are written by solo, male scholars.

In Figure S4 we look at three types of scholars: those who publish exclusively in higher-ranked journals, those who publish exclusively in lower-ranked journals, and those who publish in both. On average, nearly half of the articles produced by scholars publishing only in lower-ranked journals include an acknowledgment section, compared to approximately 63% of articles by scholars publishing only in higher-ranked journals or across both types of venues.

Finally, in Fig. S5 we examine the relationship between journal prestige and scholars’ connectivity in the acknowledgment network. Authors who publish in higher-ranked journals constitute roughly half of the population of isolated scholars. Connected scholars are more evenly distributed across journal prestige levels.

4. Evaluation of Possible Journal Effects and Uncertainty in Network Observations

As we discuss in the main paper, the acknowledgment network we analyze is a proxy to an underlying, unmeasured structure that is likely only partially represented in our measurements. Authors that seem isolated may, in fact, be connected to the informal structure. One reason for their apparent isolation may be that their articles get published in journals with a norm not to include acknowledgment notes. If those journals are also where lower quality articles are published, a ‘journal effect’ may explain the associations we report in the paper, not embeddedness (or lack of) in the informal network.

In Fig. S6 we report the association between journal prestige (according to the SCImago and CiteScore measures) and the presence of acknowledgments in the articles published. This figure shows that while papers with acknowledgments tend to be published in higher impact journals, papers without acknowledgments are also placed in highly ranked journals.

In Fig. S7 we show the results of a fixed-effects generalized linear model estimated with a binomial family to calculate the probability that an article includes an acknowledgment section. We use author-id as the fixed effect to compare authors’ behavior across journals of different prestige. The two measures of journal prestige (SCImago and CiteScore) are the independent variables. While both measures of journal prestige have a positive impact on the probability that an article has an acknowledgment section, the overall predictive power of this variable is low (i.e., the models explain about 20% of the variation in the data). However, this association is still compatible with the possibility of confounding and ‘journal effects’ in the inclusion of acknowledgment notes.

In order to address this possibility, we replicate the analyses shown in Fig. 3 of the main text adding weights to the regression models that account for imperfect observations and possible journal effects. Specifically, we extract baseline probabilities for each journal to include an acknowledgment section using the same fixed-effects generalized linear model used for figure S7 (i.e., including authors fixed effects). Using these baseline probabilities, we construct two alternative weighting schemes that assign greater relevance to observations for which we have higher confidence in their isolated status.

First, we implement a frequentist approach defined as:

$$P(T = 0) = \prod_{i=1}^n (1 - p_i), \quad [1]$$

where p_i denotes the probability that the publication venue of paper i includes an acknowledgments section. This expression captures the likelihood that none of the scholar’s publications contain acknowledgments, given the baseline propensities of the journals in which the papers are published.

Second, we adopt a Bayesian mixture-model approach where the probability of having a tie despite not observing any acknowledgment tie is defined as:

$$P(T = 1 \mid A = 0, X) = \frac{\pi(X) \cdot \prod_{i=1}^n (1 - p_i)}{\pi(X) \cdot \prod_{i=1}^n (1 - p_i) + (1 - \pi(X))}, \quad [2]$$

where $\pi(X)$ is the *prior probability of being connected*, defined as a function of author characteristics X , including gender, seniority, geographical location, and methodological orientation. The term p_i again represents the baseline probability that the venue of paper i includes an acknowledgment section. Because our analyses rely on a dichotomous classification (isolated vs. non-isolated), we assign a weight of 1 to scholars with observed connections, and use the corresponding estimated probabilities to down-weight scholars classified as isolated. Fig. S8 shows the two weight distributions.

In Figs. S33 and S34 we replicate the results of Fig. 3 in the main text with the weights. The results confirm that our main findings are robust to correction for missing informal connections. Across specifications, the estimated effect of informal academic ties remains positive and statistically significant. Further, the magnitude of the estimated coefficient of informal ties is always statistically larger than that of formal ties with only one exception: the matched-data generalized linear model with the Euclid Score as the DV and frequentist weights. When we incorporate frequentist weights, the magnitudes reverse although they are not statistically distinguishable from each other. When we incorporate Bayesian weights the magnitude for informal ties is estimated to be larger than that of formal ties, although again the difference is not statistically significant. The random forests models (which used all data) are all consistent with our main message. These results confirm the robustness of our findings even when we take into account uncertainty in our observations.

5. Global Measures of Academic Impact

We collected additional metadata for the authors of the articles, including yearly number of publications, annual citation count, and country of institutional affiliation (see Fig. S9 for the global map distribution of scholars in our sample). For each author we also obtained basic metadata on articles published in journals other than those in our list (i.e., out-of-sample publications). In the main text we use the h-index and Euclid scores calculated from the articles in our sample (i.e., articles published in the journals listed in S1). Here, we reproduce the analyses using the global h-index and Euclid scores (i.e., based on all publications associated to each scholar, see Fig. S10 for the distribution). In Fig. S11 and Fig. S12 we show the gaps in publication impact according to these global measures using the 2021 and 2018 networks to define the groups, respectively. These figures confirm that connected individuals have, overall, a higher publication impact regardless of the metric used or the decile (defined, again, using the within-field productivity distribution). The main difference is that scholars in the last productivity decile have higher global h-index and Euclid scores. These scholars are likely to be very prolific in other fields, with occasional publications in

122 Political Science journals. This explains why, when using global impact metrics not constrained by our sample, the patterns
123 across deciles reverse (compared to Fig. 2 in the main text).

124 6. Methodological Orientation

125 We assigned articles to three methodological categories: quantitative, qualitative/normative, or formal theory (see Table S5
126 for the fraction of articles in each category by journal). We implemented a two-stage supervised pipeline to perform this
127 classification. Our training set consisted of 1,694 manually coded papers from ref. (1). The papers in this original source were
128 labeled using six categories: statistical (1,138), experimental (145), qualitative (140), formal theory (118), political theory
129 (105), and conceptual (48). We first consolidated these labels into three categories: quantitative (1283), qualitative/normative
130 (293), and formal (118). Full texts were cleaned (i.e., we removed the preamble, bibliography, short or low-text lines, and
131 stop-words) and vectorized with TfidfVectorizer. We used a one-vs-rest logistic regression to fit one classifier per class. Initially,
132 we limited the vocabulary of the model to 450 features, including both unigrams and bigrams. In stage two, we took the 70
133 most significant features from the initial classifiers and retrained a model with this limited vocabulary to reduce noise in the
134 text features. A full classification report using a 10 fold cross validation and a list of coefficients is shown in Tables S2 and S3.

135 Once the final classifiers were trained, we used them to label our entire sample of full-text papers into two categories:
136 ‘quantitative/formal theory’ and ‘qualitative/normative’. We then identified the predominant category at the scholar level: if
137 an author had $\frac{2}{3}$ of their articles labeled as ‘quantitative/formal theory’ or ‘qualitative/normative’, we assigned those labels
138 (respectively); if they did not reach that cutoff, the category we assigned is ‘mixed methods’.

139 7. Institutional Prestige

140 Our data has 12,439 unique affiliations to institutions, departments, and research centers. To obtain information about
141 the academic prestige of these institutions, we matched these affiliations with three academic rankings: (1) the Academic
142 Ranking of World Universities (ARWU), also known as the Shanghai ranking; (2) the QS World University Rankings; and (3)
143 the Times Higher Education World University Rankings, or the THE rankings. We first standardized institutional names,
144 grouping departments and schools into their parent institution. Of the three listings, the Shanghai ranking returned the highest
145 percentage overlap: it allowed us to retrieve ranking information for a total of $N = 761$ institutions covering 47,619 scholars’
146 affiliations (55.59% of the total in our data, see Fig. S13).

147 Given the large number of unmatched cases, we operationalized the measure of institutional prestige as a binary variable
148 that tracks whether a scholar is in a top 100 institution, according to the Shanghai ranking.

149 8. Impact of Filtering Threshold

150 As we explain in the main paper, we filtered authors with fewer than 10 publications in order to have enough data to calculate
151 the in-sample measures of publication impact. Here we offer some descriptive analysis of the impact this filtering threshold has
152 on the distribution of the data. In section 16 we also offer additional analyses to determine how using a different threshold (5
153 publications) or no threshold at all alters our main results. The summary of these tests is that removing the threshold adds
154 more noise to the data, which worsens the performance of the models, but the overall conclusions reported in the main text
155 hold.

156 In Fig. S14 we show the impact that the productivity threshold has the gender, seniority, geographical region, and
157 methodological orientation of the scholars in the data. The threshold affects mostly the presence of senior scholars: after the
158 10-paper cut-off, the ratio of senior to junior authors is significantly skewed. The imbalances are not so stark for the other
159 variables.

160 In Fig. S15 we offer more detail on the gender breakdowns by methodological orientation. After the threshold, there is a
161 slightly larger methodological imbalance, with more scholars of a quantitative orientation relative to qualitative scholars. The
162 gap, however, is similar regardless of gender (top row for female, bottom row for male).

163 The gap between male and female scholars can be seen across all deciles of the productivity-within-the-field distribution, as
164 shown in Fig. S16. This gap is accentuated after the 10-article filter, especially in the first two deciles.

165 In Fig. S17 we confirm that the filter excludes a higher fraction of junior scholars. Those who remain after the filter tend to
166 do more quantitative work. See section 16 for additional sensitivity analyses.

167 9. Network Fragmentation

168 To determine if the fragmentation observed in the two networks is larger than expected by chance, we compare the networks
169 with 1,000 random permutations that preserve the same size, density, and degree distribution, also known as the configuration
170 model (2). Fig. S18 shows the result of the test for the co-authorship network. Fig. S19 shows the result of the test for the
171 acknowledgment network. In both instances, the observed level of fragmentation is substantially higher than it would be in a
172 network with the same structural characteristics but randomly created connections.

173 10. Network Overlap

174 Table 1 in the main text offers basic descriptive statistics of the two networks we analyze. The table shows that the
175 acknowledgment network is significantly larger, with an average degree that is twice as high as that of the co-authorship

network. It also shows that the acknowledgment network is substantially less fragmented (see figures S18 and S19 as well). Here we discuss additional descriptive data to show that scholars tend to hold different local positions in the two networks.

In Fig. S20 we show the scatterplot of the degree centrality for scholars that are present in both networks. The top row is based on the full dataset, the bottom row uses the filtered data (which requires scholars to have 10 or more published articles, see also section 8). Panels A and C show that the correlations are weak with and without the 10-paper filter. Panels B and D show the correlation coefficients within deciles. The strongest alignment happens in the top decile, i.e., the more central scholars in the co-authorship network tend to be more central in the acknowledgment network. The correlation coefficients, however, are still moderate in magnitude.

In table S6 we further split the degree range, average, and median by gender. Male and female scholars hold similar positions in the co-authorship network (with female authors slightly better connected, on average). In the acknowledgment network, there are more male outliers in the degree distribution: male scholars hold, on average, 2.5 more connections than female scholars.

11. The Impact of Network Centrality

In Fig. 4 of the main text we show that centrality in the acknowledgment network is the most important predictor of academic impact. In this section we offer additional descriptive plots to assess the relationship of network centrality and impact.

In Fig. S23 we show the first-order correlation of degree with our two measures of impact: the h-index (top row) and the Euclid scores (bottom row). The association is moderate in both cases, but stronger in the case of the acknowledgment network, in line with our main finding that informal connections matter more to explain variation in impact. The correlation for female scholars is slightly lower in all cases than for male scholars.

In Fig. S24 we display the correlation between network centrality and publication impact for each decile of the within-field productivity distribution, with and without the 10-paper threshold (see also section 8). Here we see that, absent the 10-paper filter (panels A-D), there is more noise in the subset of authors classified in the first two deciles. Without the filter, a scholar with only one article, published in a political science journal, will be in the same decile as a scholar that has 11 articles, all published within the field. The second scholar, however, offers more data points to assess their overall publication impact. When we restrict the analysis to scholars with at least 10 publications (panels E-H), we remove these noisier observations.

In Section 15 we offer additional analysis that extend the results reported in Fig. 4 of the main text. The noise-reduction effects of applying the 10-paper filter can also be seen in the performance of the models discussed in Section 16.

12. Network Homophily

Network homophily describes the tendency of actors to build connections with similar others (2, 3). In social networks, this mechanism is often summarized with the proverb ‘birds of a feather flock together’, or the principle that similarity breeds connection (4). Here we analyze this tendency in the two networks around the four attribute categories we use as predictors in our models: gender, institutional prestige, seniority, and country of affiliation.

We define the homophily of a node i with an attribute A_i as:

$$H = \frac{\sum_{(i,j) \in E} \mathbf{1}[A_i = A_j]}{|E|}$$

where E is the set of edges in the network and A is the categorical attribute of interest. This measure captures the fraction of ties that connect nodes with identical attributes.

In Fig. S25 we plot this measure. In particular, we plot the number of scholars that have x fraction of connections to similar others. When this fraction equals 1, all alters have the same attribute as ego, when it is 0, the alters are all different. The distributions suggest that the acknowledgment network has a more diverse composition across all attributes.

Shifting attention to paper-level analyses, Fig. S26 shows that female-only teams write the highest fraction of articles that acknowledge other female scholars, with all-male author teams are the least likely to acknowledge females.

These observed levels of homophily result from two underlying factors: compositional constraints (e.g., there are so many female scholars to connect to) and selection (e.g., male scholars choose to connect with other male scholars). To differentiate these two factors, we compare the observed levels of homophily with what would be expected by random chance, given compositional constraints. We generate two random benchmarks: (1) one derives from 1,000 permutations of the network following the configuration model (2); (2) the second derives from holding the network constant but randomly relabeling the node attributes ($\times 1,000$). We then use z -scores to measure the distance between the observed homophily (red dashed lines in Figs. S27 and S28) and the random expectation, given the compositional characteristics of the data (blue histograms). Following past research (5, 6), we interpret the magnitude of this distance as the strength of the selection bias, given the compositional homophily already intrinsic in the data.

In all instances, homophily is larger than expected by chance, but especially so for country of affiliation and seniority level (with junior scholars connecting to other junior scholars more frequently than their presence in the data would suggest). The z -scores tell us that gender homophily is, overall, stronger for females across the two networks.

In Figs. S29 and S30, we display similar analyses but with an alternative measure of homophily: assortative mixing. Assortativity provides a graph-level measure of similar-node attachment. It generalizes the idea of homophily by quantifying the correlation between node attributes across all connected pairs (7). A positive assortativity coefficient indicates that similar

nodes tend to connect more often, whereas a negative value signals disassortative mixing. Values close to 0 suggest random mixing in the network.

The assortativity coefficient in a graph partitioned in groups i and j according to some attribute is defined as:

$$r = \frac{\sum_i e_{ii} - \sum_i a_i b_i}{1 - \sum_i a_i b_i}$$

where e_{ij} is the fraction of edges that connect nodes of type i to nodes of type j , and $a_i = \sum_j e_{ij}$ and $b_j = \sum_i e_{ij}$ are the corresponding row and column marginals.

Country of affiliation emerges, again, as the most important clustering factor in both networks.

13. Correlation Matrix

In Fig. S31 we show the pairwise correlation coefficients for all non-binary variables considered in the regression and classifier models. The number of articles published in the journals we track is strongly correlated with our measures of publication impact (unsurprisingly, since it is an important component of the operationalization of both measures of impact). The gender, seniority, and country homophily measures (see section 12) are also correlated within and across networks. Our two main explanatory variables (degree in the co-authorship network and degree in the acknowledgment network) are only weakly to moderately correlated with all the other variables.

14. Coarsened Exact Matching Regression Models and Threats to Inference

In Fig. 3 of the main text we show that connectedness in the acknowledgment network correlates strongly with academic impact. In Fig. S32 we show that it remains the strongest predictor across model specifications (see also Tables S7-S10 for full regression outputs).

In the absence of unobserved confounders that influence both network placement and academic impact, the differences we observe can be interpreted as causal. However, we cannot observe all the factors that shape network embeddedness and academic impact. While we cannot randomize network placement — nor would we want to, given our interest in the endogenous structure of scholarly networks — we address the challenge of confounders with different strategies. First, we alleviate the concerns related to confounders by reweighting observations to ensure that connected and disconnected scholars are as similar as possible along the observable covariates: gender, location, seniority, institutional prestige, and methodological orientation. We focus on these variables because past work suggests they are significant correlates of academic impact, e.g., (8–10). If connected and disconnected scholars differ only across these observable characteristics — the most salient in the extant literature — this strategy is sufficient to retrieve the difference attributable to connectedness alone.

In addition, we modified the definition of group boundaries (connected/disconnected) in different ways to test the robustness of our findings while applying different constraints to our comparisons. In Fig. S21 we reproduce the impact gap analysis using the networks measured 5 years prior to the measures of impact. As discussed in section 5, Figs. S11 and S12 use global measures of impact and networks built 2 and 5 years prior to measuring impact. In Fig. S22, we evaluate the gap in the subset of the data that includes only scholars who are connected in the co-authorship network. And, finally, in Fig. S33 (and Tables S11-S18) we show the results of re-estimating the models using frequentist and Bayesian weights capturing measurement uncertainty (see also section 4). Across all these specifications, results are consistent with our main conclusion that informal ties matter, i.e., they help explain variation in academic impact.

Figure 4 of the main text, and the additional robustness and sensitivity analyses presented in sections 15 and 16, show that connectedness in the acknowledgment network remains, overall, the most important predictor also when operationalized as a continuous variable in the form of degree centrality. The consistency of findings across all these different constraints and specifications strongly suggests that connectedness itself, not characteristics correlated with it, drives our results.

15. Random Forest Regressions and Classifiers

In addition to the random forest regressions reported in Figs. 3 and 4 of the main text, we run additional models to test the robustness of the findings to different model specifications. Fig. S34 shows the results of re-estimating the random forest model shown in Fig. 3 of the main text with the frequentist and Bayesian weights (see section 4). Another set of analyses includes running tests with a categorical operationalization of our DVs, i.e., the h-index and Euclid scores. We split these measures of publication impact in their four quartiles and define the problem as one of classification. A third set of analyses includes using different algorithms to the classification task. Across all these specifications, the results are consistent with our main findings, i.e., informal ties remain the most relevant predictor of academic impact.

Random forest classifier Similarly to the Random Forest Regressor discussed in the main text, the Random Forest Classifier (RFC) evaluates several decision trees and averages the resulting categorical output (11). We first fine-tuned the RFC hyperparameters by means of a grid search cross validation that performed an exhaustive search of hyperparameters to select the best ones given the chosen predictor. We evaluated the performance by means of a 10-fold cross validation, i.e., the dataset is split in 10 folds and each one is used iteratively as test data while the rest are used as training data until each fold has been test-used.

XGBoost classifier This classifier offers a variation of a gradient boosting algorithm that sequentially designs an ensemble of decision trees trying to minimize the loss function with the errors of precedent trees (12). Compared to other gradient boosting algorithms, XGB takes a second order approximation to the loss function allowing a better handling of complex data.

Support vector classifier This version of the classifier is based on Support Vector Machine learning models, which are widely used algorithms in machine learning (13). These algorithms assess the optimal hyperplanes to maximize the separation between classes in a given sample. When the data cannot be separated in a linear manner, the algorithm finds higher dimensional planes trying to balance between a good separation between classes and minimizing the classification errors.

Training parameters Hyperparameters for classification and regression models were optimized using a grid search with stratified 3-fold cross-validation. The final configurations were:

- Random Forest Regressor: max depth = 15; min samples leaf = 2; min samples split = 5; max features = $\log_2$ (of the number of predictors); n estimators = 150;
- Random Forest Classifier: max depth = 15; min samples per leaf = 2; min samples split = 5; max features = $\log_2$ (of the number of predictors); criterion = entropy.
- XGBoost Classifier: number of estimators = 50; max depth = 5; $\alpha = 0$ (no L1 regularization); $\gamma = 0$ (no minimum loss reduction); learning rate = 0.1; subsample = 0.8.
- Support Vector Classifier: $C = 1$; $\gamma = \frac{1}{n_{\text{features}}}$ (scikit-learn’s auto setting); class weight = balanced, kernel = rbf.

Feature importance: Binary Variables. Figs. S35, S36, and S37 are extensions to Fig. 3 in the main text. Predictive accuracy is strongest for the first and last quartiles of the publication impact distribution, and in general the models perform better for the h-index measure.

We define variable importance as follows: for each fold of the classification algorithm, we obtain the feature importance (FI) by permutation error increase for each independent variable v ; that is, we assess how the performance of the RFC changes when permuting the order of one variable according to a selected metric. FI is defined as follows:

$$FI(v) = s - \sum_{n=1}^N s_{n,j}(D_{f,v})$$

where v is the tested variable, s is the reference score, $s_{n,j}$ the score function, N the number of repetitions, D is the new dataset with variable v randomized in fold f . To obtain the overall FI, we average the importance for each variable across folds.

Feature importance: Count Variables. Figs. S38, S39, and S40 are extensions to Fig. 4 in the main text. Here, the models include the degree centrality in the two networks and additional variables tracking attribute homophily. These models consistently show that the number of connections in the acknowledgment network are the most important predictor of publication impact, as measured by the h-index. The models that explain variation in Euclid scores perform worse (partly because of the nature of the scores, which over-emphasize hit publications) but, in general, acknowledgment ties are also the most important feature.

SHAP values. To complement the random permutation importance, a global measure, we calculated SHAP (SHapley Additive exPlanations) values (14). SHAP is based on cooperative game theory and attributes to each feature j the average marginal contribution it makes to the model output $f(x)$ across all possible subsets S of the remaining features:

$$\phi_j(f, x) = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|! (|N| - |S| - 1)!}{|N|!} \left[f_{S \cup \{j\}}(x_{S \cup \{j\}}) - f_S(x_S) \right] \quad [3]$$

where N is the full set of input features, x_S denotes the values of the features in subset S , and ϕ_j is the SHAP value for feature j for instance x . The sum of all ϕ_j plus the expected model output equals $f(x)$, ensuring additivity.

For tree-based models (Random Forest, Gradient Boosted Trees) we used the exact *TreeExplainer* implementation in the SHAP package, which exploits the structure of the tree ensemble to compute ϕ_j efficiently. For the non-linear Support Vector Classifier we applied the model-agnostic *KernelExplainer* with a random subset of the training data as background to approximate SHAP values. Individual SHAP values were then aggregated across the test set to derive global feature-importance rankings.

16. Sensitivity Analyses

To determine the impact of the 10-paper threshold on our results, we run the random forest regression and classifiers with a dataset that lowers the threshold to 5 papers (Fig. S41) and a dataset that does not apply any thresholds (Fig. S42). These analyses show that the performance of the models worsens but, overall, the results remain qualitatively similar. See also section 8.

Table S1. List of political science journals included in the study

Electoral Stud.	SocSci. Qtrly.	Human Rights Qtrly.	Governance	New Pol. Econ.
Pub. Choice	Int'l. J. of PubOp. Rsrch.	Pol. & Society	Pol. Communication	J. of Peace Rsrch.
J. of Theoretical Pol.	Pol. Theory	Nations & Nationalism	West Euro. Pol.	Pol. Behavior
PubAdmin.	Europe-Asia Stud.	Latin Am. Perspectives	J. of Euro. PubPol.	PS - PoliSci & Pol.
Comp. Pol. Stud.	African Affairs	Int'l. PoliSci Rev.	Politicka Ekonomie	Policy & Pol.
J. of Cflct. Resolution	J. of Democracy	British J. of PoliSci	Comp. Pol.	Prblms. of Post-Communism
Pol. Geography	New Left Rev.	Am. PoliSci Rev.	J. of Pol.	Euro. J. of Pol. Econ.
Econ. & Pol.	Democratization	Communist & Post-Comm. Stud.	Citizenship Stud.	Int'l. Stud. Qtrly.
Stud. in Comp. Int'l. Dev.	Local Gov. Stud.	Geopolitics	Nationalities Papers	Gov. & Opposition
Armed Forces & Society	East Euro. Pol. & Societies	Am. Pol. Rsrch.	Int'l. Affairs	J. of PubPol.
J. of Int'l. Relations & Dev.	Australian J. of PoliSci	Euro. J. of Pol. Rsrch.	J. of Pol. Ideologies	Latin Am. Pol. & Society
Post-Soviet Affairs	PoliSci Qtrly.	Pol. Rsrch. Qtrly.	Independent Rev.	J. of Strategic Stud.
Annals of Am. Acad. of P...	Pol. Stud.	Int'l. Security	PubOp. Qtrly.	Int'l. Pol.
Scandinavian Pol. Stud.	World Pol.	J. of Common Market Stud.	Am. J. of PoliSci	Canadian J. of PoliSci
Int'l. J. of Press/Pol.	Pol. Psychology	Env. Pol.	J. of Pol. Phil.	Polity
Rev. of African Pol. Econ.	J. of PubAdmin. Rsrch. &...	Mediterranean Pol.	Scottish J. of Pol. Econ.	Stud. in Am. Pol. Dev.
Party Pol.	Int'l. Organization	Osteuropa	Pol., Phil. & Econ.	Publius: J. of Federalism
Historical Materialism	Pol. Qtrly.	Int'l. Stud. Rev.	Australian J. of Pol. & Hist.	Rev. of Policy Rsrch.
Policy Stud. J.	Stud. in Cflct. & Terrorism	Cooperation & Cflct.	Survival	PoliSci
Perspectives on Pol.	Ethics & Int'l. Affairs	J. of Policy Hist.	Rev. of Int'l. Pol. Econ.	Parliamentary Affairs
Terrorism & Pol. Violence	Phil. & Pub. Affairs	South Euro. Society & Pol.	Int'l. Env. Agreements: ...	Int'l. Feminist J. of Pol.
Swiss PoliSci Rev.	British J. of Pol. & Int...	Socio-Economic Rev.	Japanese J. of PoliSci	Euro. Union Pol.
Pol. Analysis	State Pol. & Policy Qtrly.	Forum (Germany)	Business & Pol.	Legislative Stud. Qtrly.
Critical Rev.	Euro. Hist. Qtrly.	Curr. Hist.	Policy & Society	Politics
Pol. Stud. Rev.	Intelligence & NatSec.	J. of Women, Pol. & Policy	Rev. of Int'l. Orgs.	Presidential Stud. Qtrly.
Politikon	Qtrly. J. of PoliSci	Pol. & Gender	British Pol.	Global Env. Pol.
Acta Politica	Euro. PoliSci	J. of Contemp. Euro. Stud.	J. of Euro. Integration	Int'l. Pol. Sociology
Regulation & Governance	J. of Chinese PoliSci	J. of Info. Tech. & Pol.	J. of Australian Pol. Econ.	Ethics & Global Pol.
Comp. Euro. Pol.	Contemp. Pol. Theory	Irish Pol. Stud.	Euro. PoliSci Rev.	Lex Localis
Pol. & Religion	Int'l. Theory	Contemp. Security Policy	Social Movement Stud.	Contemp. Pol.
J. of Human Rights	Global Policy	Critical Policy Stud.	Revista Brasileira de Po...	German Pol.
J. of Elections, PubOp. ...	Cambridge Rev. of Int'l....	Telos	Int'l. J. of Cflct. & Violence	Politix
Euro. Security	Romanian J. of PoliSci	Int'l. J. of Transitiona...	J. of Cold War Stud.	Policy & Internet
Contemp. Southeast Asia	East Euro. Pol.	Territory, Pol., Governance	Pol. & Governance	Rsrch. & Pol.
J. of Curr. Southeast As...	PoliSci Rsrch. & Methods	J. of Chinese Governance	Peacebuilding	

Table S2. LogReg method classification report

Category	Precision	Recall	F1-score	Support
Quantitative	0.93	0.96	0.95	1283
Formal	0.88	0.65	0.75	118
Qual/Theory	0.85	0.81	0.83	293
Micro avg	0.92	0.92	0.92	1694
Macro avg	0.88	0.81	0.84	1694
Weighted avg	0.91	0.92	0.91	1694
Samples avg	0.92	0.92	0.92	1694

Table S3. Top 40 words by classification coefficient for each method category

Quant		Formal		Qual/Norm	
equilibrium	-3.89	equilibrium	5.57	model	-2.66
variables	2.91	model	2.53	results	-2.56
results	2.86	game	1.93	effect	-2.42
variable	2.83	bargaining	1.79	variable	-2.30
data	2.57	cost	1.78	variables	-2.19
effect	2.38	data	-1.52	justice	1.77
respondents	2.11	political	-1.41	social	1.76
likely	2.09	probability	1.30	interests	1.73
case	-1.98	variables	-1.28	data	-1.73
table	1.94	rule	1.09	new	1.67
attitudes	1.77	variable	-1.05	causal	1.63
effects	1.73	respondents	-1.02	respondents	-1.62
sample	1.72	war	1.01	good	1.59
good	-1.64	significant	-1.00	likely	-1.58
new	-1.62	subjects	-0.97	attitudes	-1.55
example	-1.59	democratic	-0.96	did	1.54
social	-1.57	table	-0.94	probability	-1.49
measure	1.51	effects	-0.93	theory	1.42
causal	-1.50	case	0.93	equilibrium	-1.42
cost	-1.44	type	0.90	table	-1.41
using	1.43	public	-0.90	sample	-1.40
subjects	1.39	network	0.88	people	1.39
strategy	-1.37	results	-0.88	way	1.38
justice	-1.36	target	0.87	human	1.38
estimates	1.36	control	-0.85	effects	-1.35
ndings	1.31	treatment	-0.84	like	1.32
statistically	1.28	beliefs	0.82	liberal	1.28
interests	-1.27	strategy	0.82	state	1.25
way	-1.26	lower	0.81	society	1.22
significant	1.26	people	-0.80	power	1.22
significant	1.24	equation	0.78	market	1.20
theory	-1.23	result	0.76	organizations	1.19
impact	1.22	elections	-0.76	example	1.19
rule	-1.19	investment	0.76	reform	1.19
levels	1.16	women	-0.76	view	1.17
make	-1.15	partisan	-0.75	law	1.16
panel	1.14	terrorist	0.74	scholars	1.16
year	1.14	using	-0.74	crisis	1.16
problem	-1.13	estimates	-0.74	problem	1.15
private	-1.12	median	0.73	case	1.15

Table S4. Acknowledgments to Research Assistants (see section 2 for more details on data extraction).

Category	Count	Proportion
Papers		
No research assistant acknowledged	119,768	0.923
At least one research assistant acknowledged	9,983	0.077
Unique research assistant names		
Female	9,881	0.483
Male	10,306	0.503
Unknown	271	0.013

```

1 Task: Extract entities from the acknowledgment section of an academic paper. Categorize them
   into commenters, reviewers, research assistants, organizations, seminars, funding sources,
   and other people. Since the data is occasionally messy, code the includesAckSection variable
   as 1 or 0 depending on whether an acknowledgment section is actually in the excerpt.
2
3 Note: Commenters includes those who contributed intellectually to the project. For example,
   this would not include someone who simply shared data, but it would include someone who
   helped make sense of data. Just sharing data does not make somebody a commenter. Commenters
   includes only those names explicitly mentioned (i.e. do not include phrases like 'workshop
   participants' or 'reviewers' or 'members of...' in the commenters category). Editors,
   reviewers, discussants, and conveners, even if mentioned by name, should not be considered
   commenters. Occasionally the authors themselves will be mentioned in the acknowledgment
   section, usually in the form 'Professor ____ is a researcher at XXX University...' In these
   cases, do not accidentally include the authors in the commenter category. Finally,
   commenters and research assistants are mutually exclusive, and so someone cannot be both a
   commenter and an assistant.
4
5 Here are several examples of acknowledgment sections and their codes (note, the real data will
   be more messy):
6
7 Input: The authors wish to thank John Smith, Jane M. Doe, and Dr. Robert Johnson for their
   valuable feedback. Earlier versions of this paper were presented at conferences at Stanford
   University and MIT. We are grateful to our research assistant, Emily Chen, and the anonymous
   reviewers for their insightful comments. This work was supported by a grant from the
   National Science Foundation.
8
9 Output:
10
11 'commenters': ['John Smith', 'Jane M Doe', 'Robert Johnson'],
12 'reviewers': ['anonymous reviewers'],
13 'researchAssistants': ['Emily Chen'],
14 'organizations': ['National Science Foundation'],
15 'seminars_or_workshops': ['Stanford University', 'MIT'],
16 'fundingSources': ['National Science Foundation'],
17 'otherPeople': [],
18 'includesAckSection': 1
19
20
21 Input: Ryan Thoreson and Andrew Clark are a PhD candidates at Yale University. We thank Jorge
   Dominguez, Ruben Dur- ante, and Tim Frye for their helpful comments. Our research assistants
   , Lisa Wang and Mark Taylor, provided excellent support. We are grateful to the anonymous
   referees and the associate editor, sally-mae gross, for their constructive feedback. This
   work was presented at the American Political Science Association Annual Meeting (
   Philadelphia) 2016 and was funded by a grant from the MacArthur Foundation.
22
23 Output:
24
25 'commenters': ['Jorge Dominguez', 'Ruben Durante', 'Tim Frye'],
26 'reviewers': ['anonymous referees', 'associate editor'],
27 'researchAssistants': ['Lisa Wang', 'Mark Taylor'],
28 'organizations': ['American Political Science Association', 'MacArthur Foundation'],
29 'seminars_or_workshops': ['American Political Science Association Annual Meeting'],
30 'fundingSources': ['MacArthur Foundation'],
31 'otherPeople': ['Sally Mae Gross'],
32 'includesAckSection': 01
33
34 -----
35
36 Now it's your turn. Extract and categorize entities from the following acknowledgment section.
   Make sure to output your response in valid json (in brackets, with double quotes, and not
   including \n strings.):
37
38 Input: {EXCERPT}

```

Fig. S1. Acknowledgment Extraction Prompt

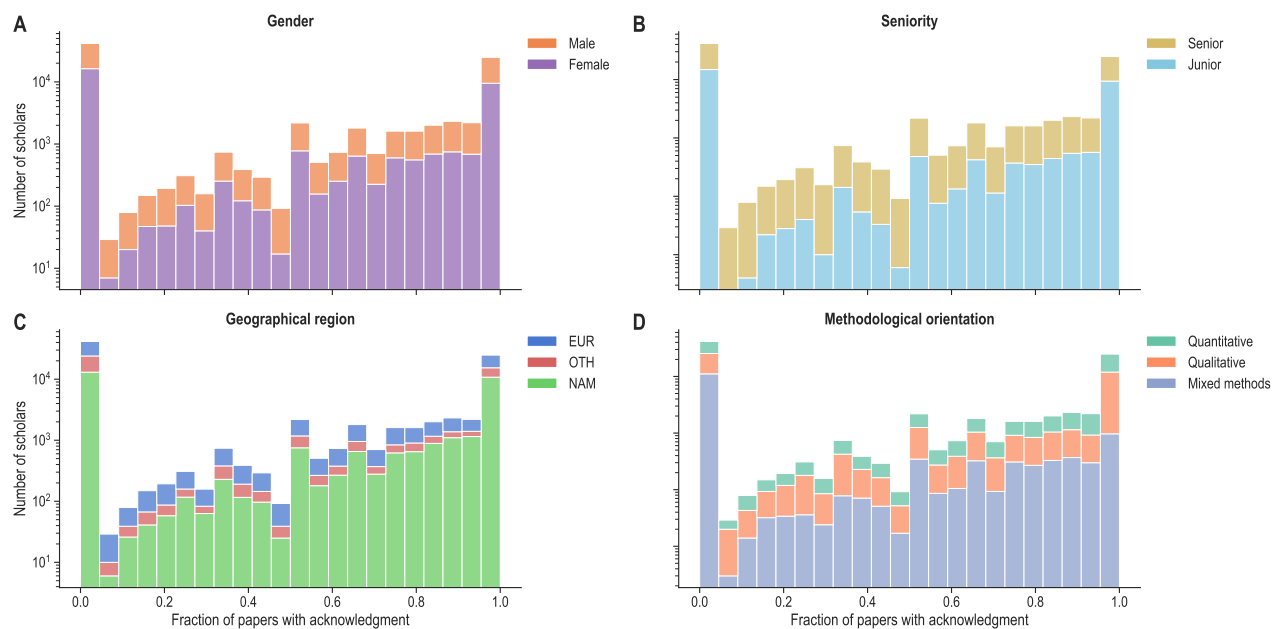


Fig. S2. Fraction of articles by scholar with an acknowledgment section. The panels plot the number of scholars that have a given x fraction of articles with an acknowledgment section. The majority of scholars have at least one article in which they acknowledge other scholars, but $\sim 42,000$ have 0 papers with an acknowledgment section. Each panel shows the same distribution split by co-variables (note that histograms within each panel are overlaid to simplify visualization): (A) shows there are more male scholars than female scholars in each bin of the distribution; (B-D) show that senior scholars based in Europe and of a quantitative orientation also prevail in each bin.

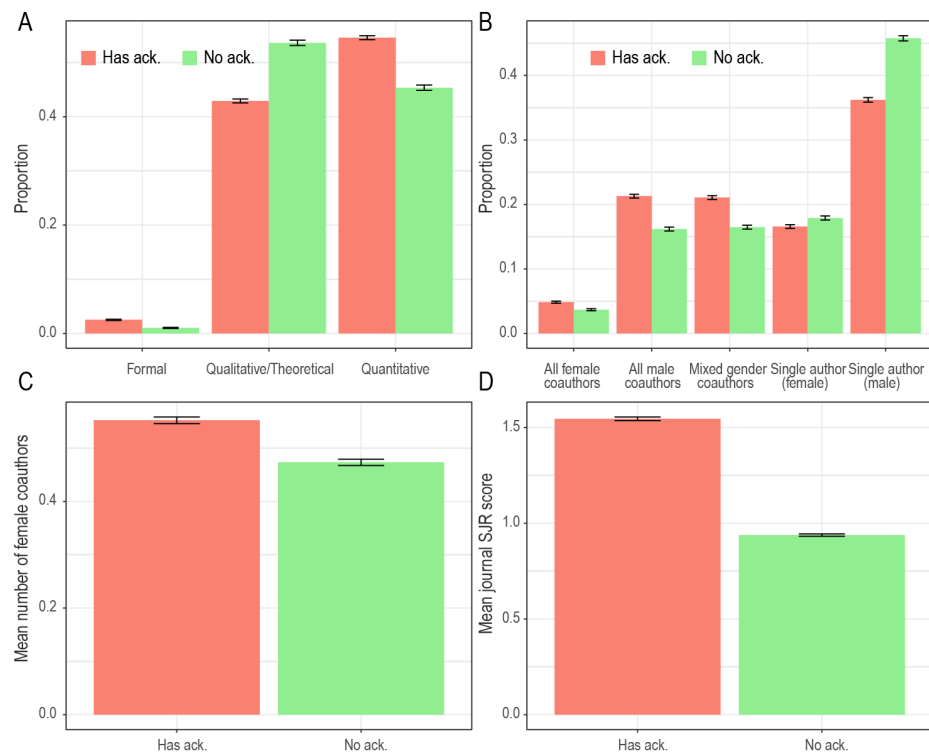


Fig. S3. Fraction of articles with an acknowledgment section. (A) Articles grouped by methodological orientation. (B) Articles grouped by authoring team composition. (C) Articles with an acknowledgment section have, on average, more female co-authors. (D) Articles with an acknowledgment section are published, on average, on more highly ranked journals.

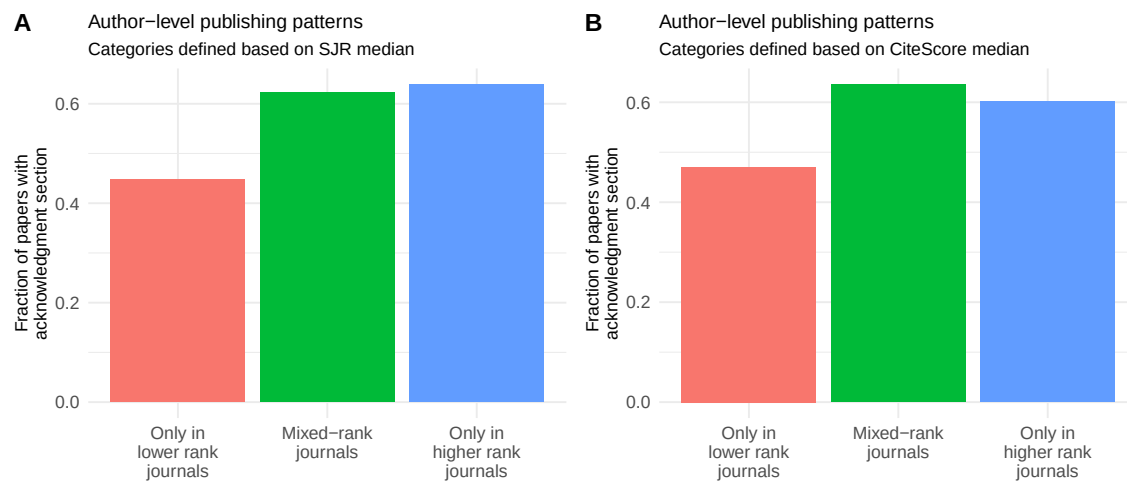


Fig. S4. Fraction of articles with an acknowledgment section by author publication profile. Fraction of articles with an acknowledgment section authored by scholars who only publish exclusively lower-ranked journals (red), higher-ranked journals (blue) or present a mixed publication pattern between both (green) according to SCImago (A) and CiteScore (B).

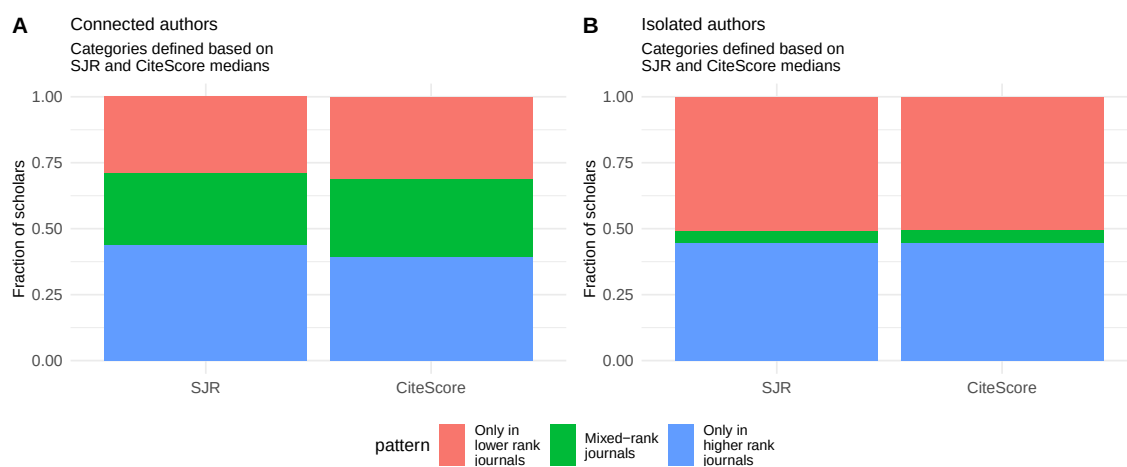


Fig. S5. Fraction of scholars by publication profile and network connectivity. Scholars are classified as publishing exclusively in lower-ranked journals (red), exclusively in higher-ranked journals (blue), or across both (green), based on journal rankings. Panels show connected (A) and isolated (B) scholars in the acknowledgment network. Isolated scholars are predominantly those publishing exclusively in either higher- or lower-ranked journals, with a smaller share publishing across both types of venues.

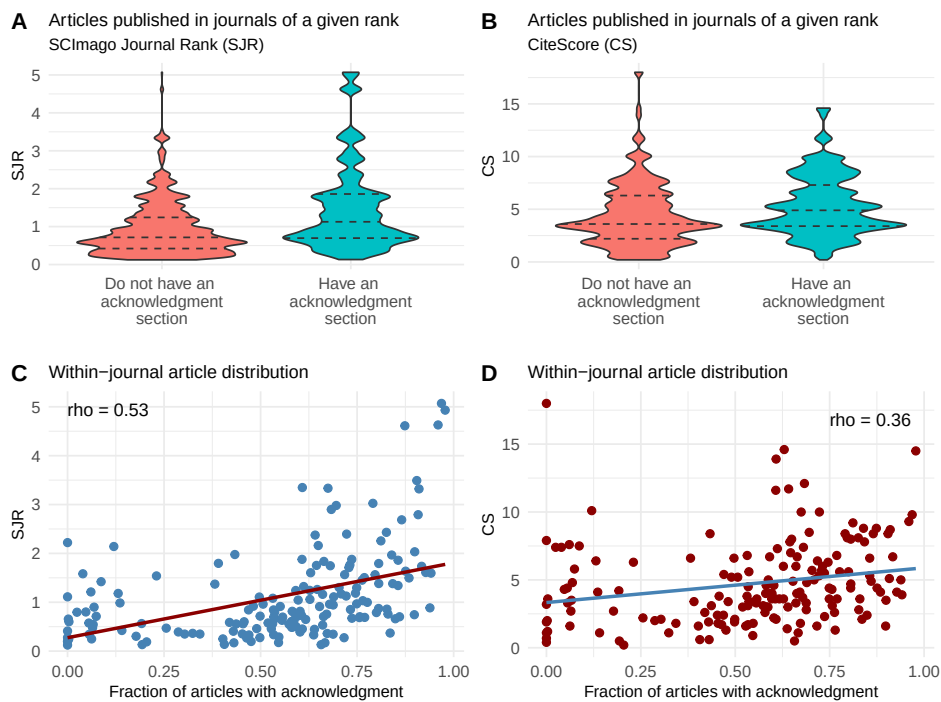


Fig. S6. Journal prestige and acknowledgment inclusion. (A) Distribution of journal prestige for articles that include an acknowledgment section and for those that do not, based on SCImago Journal Rank (SJR), and (B) CiteScore. (C) Journal prestige as a function of the fraction of articles with an acknowledgment section within each journal, where each point represents a journal. Solid lines indicate linear best-fit relationships. Overall, the association between journal prestige and acknowledgment inclusion is positive but moderate for SJR (C) and weak for CiteScore (D).

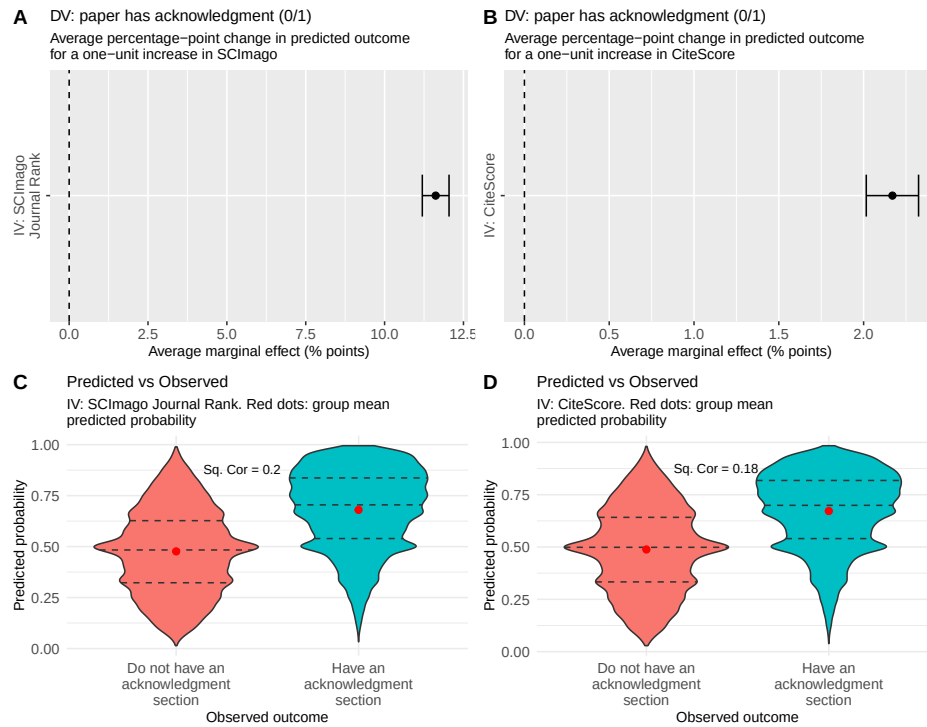


Fig. S7. Author-level fixed marginal effects. (A) Average marginal effects from fixed-effects generalized linear models using SCImago Journal Rank (SJR) and (B) CiteScore as measures of journal prestige. Horizontal error bars denote standard errors of the estimated marginal effects. (C) Predicted probability distributions of including an acknowledgment section when using SJR and (D) CiteScore as independent variables. Red points indicate the mean predicted probability within each prestige group. Overall, the predictive power of journal prestige is low, suggesting that venue ranking is not the primary determinant of whether an article includes an acknowledgment section.

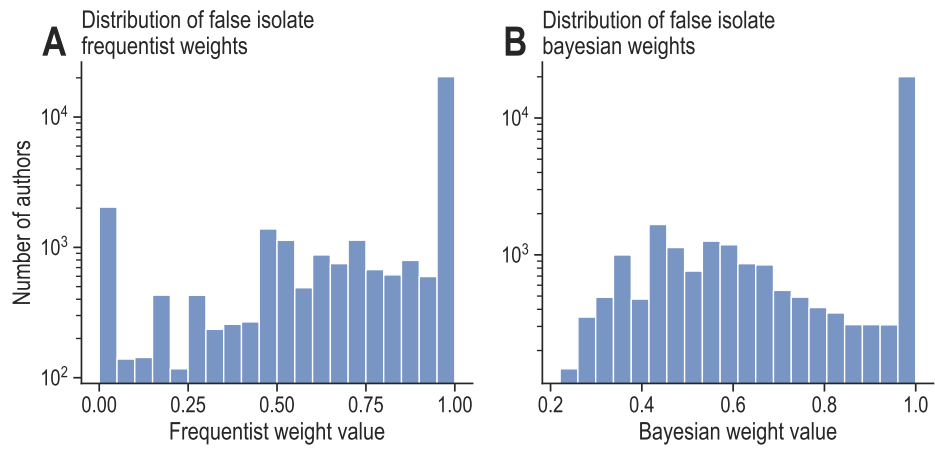


Fig. S8. Frequentist and Bayesian weight distribution. According to our definitions in section 4, all authors connected in the informal network are assigned a weight of 1 and 'isolated' authors are assigned a weight capturing measurement uncertainty, following a frequentist (A) and Bayesian (B) approach to the estimation. See also Figs. S33 and S34.

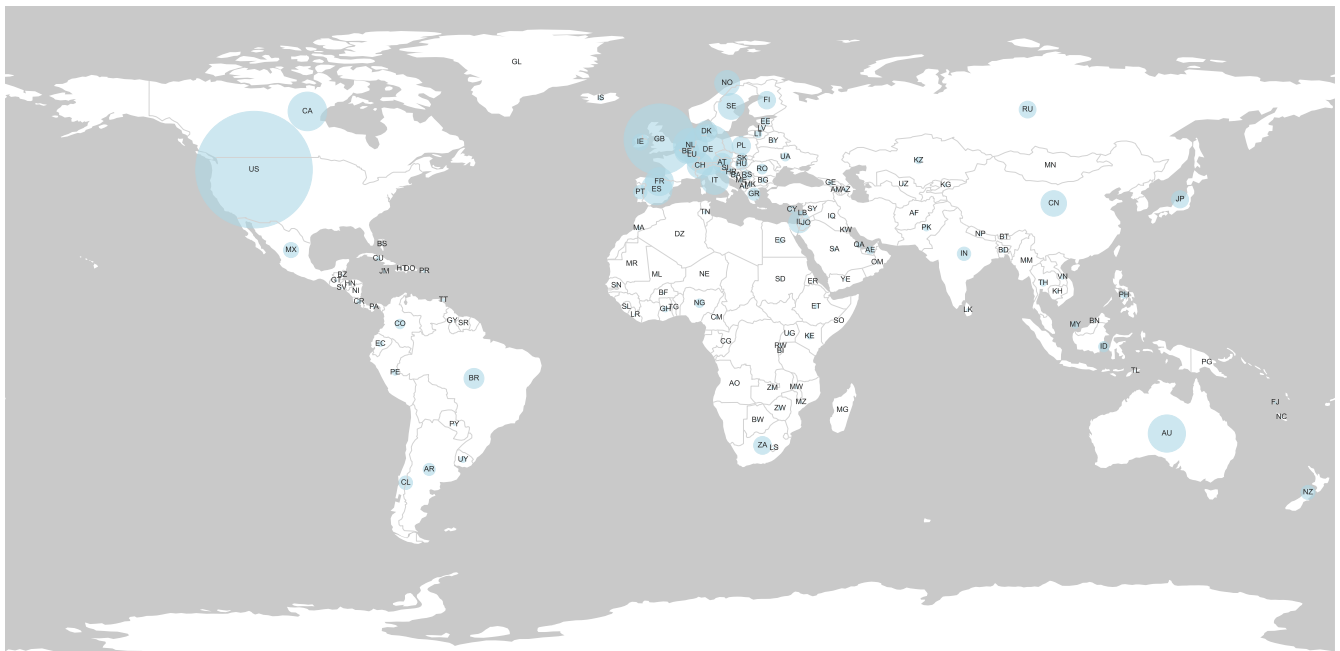


Fig. S9. Global distribution of authors. Map showing the distribution of authors based on their country of origin. The size of each bubble is proportional to the number of authors from each country. All countries with authors in the sample are indicated with the two-digit iso code. Countries in blank have no authors in our sample.

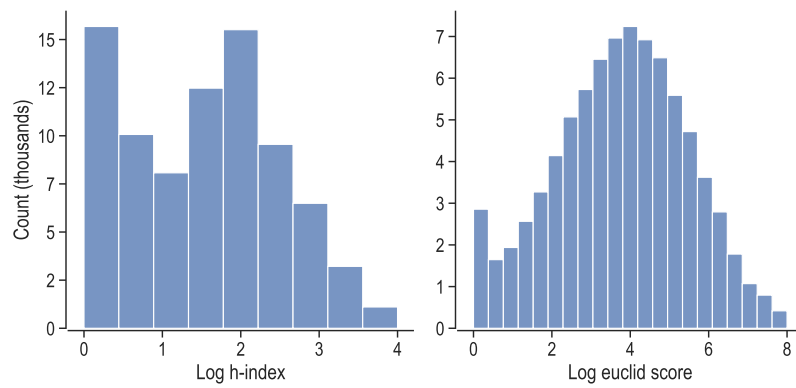


Fig. S10. Global impact metrics. Distribution of the publication impact measures for all articles published by the authors in our sample (whether they appeared in any of the journals listed in [S1](#) or not).

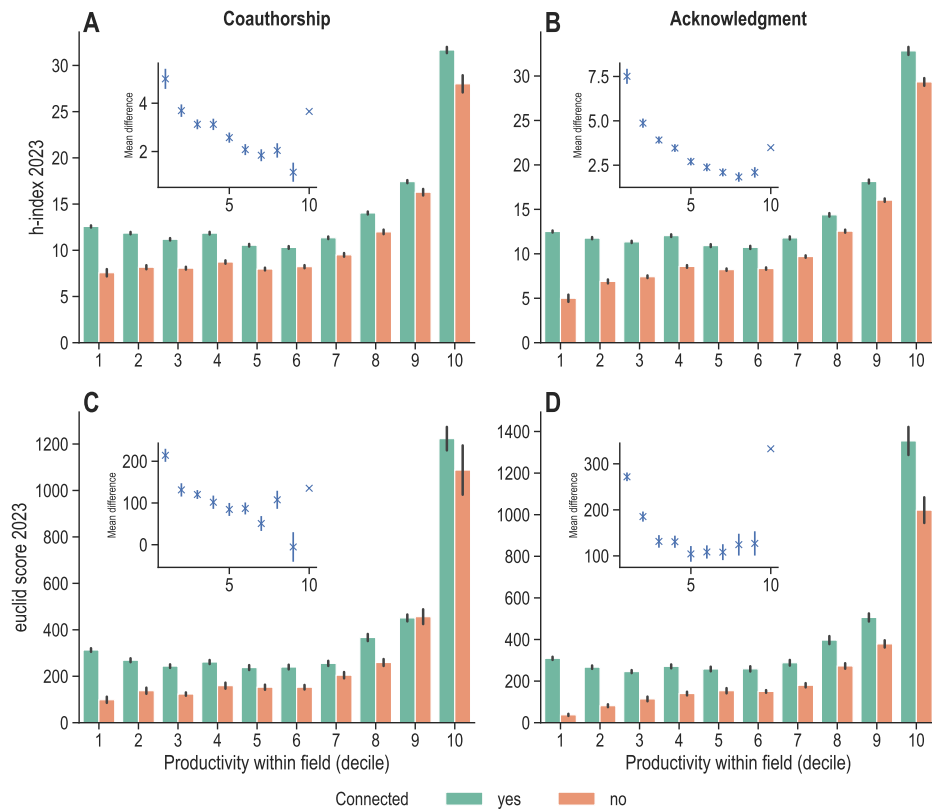


Fig. S11. Differences in publication impact. Extension of Figure 2 in the main text using the global impact metrics and the networks as they existed in 2021. Despite the distortion introduced by highly cited scholars with occasional publication in Political Science, we still observe how connected scholars have a higher impact than disconnected ones.

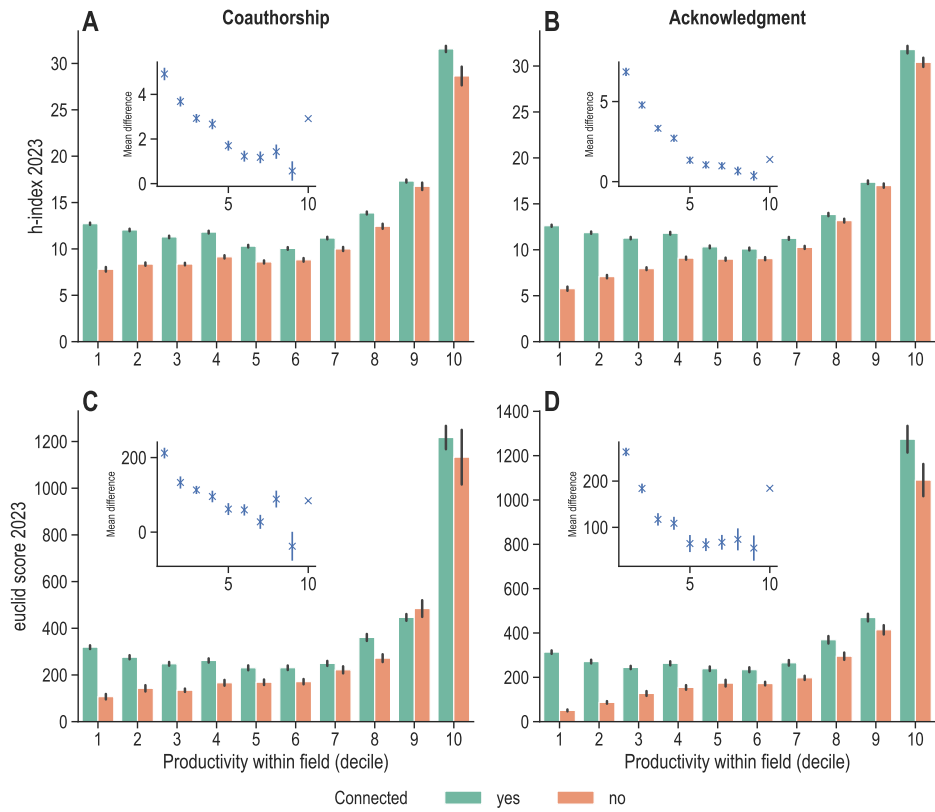


Fig. S12. Differences in publication impact. Extension of Figure 2 in the main text and to [S11](#) using the global impact metrics and the networks as they existed in 2018. The correlation of connectedness and impact is also visible when using a 5-year lag.

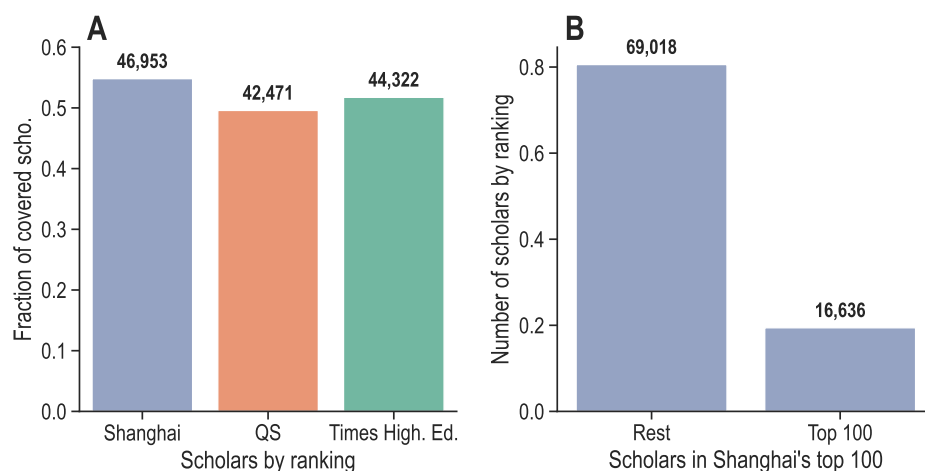


Fig. S13. Sources of academic ranking. (A) assesses the coverage of three different sources of academic rankings: the Academic Ranking of World Universities (ARWU), also known as the Shanghai Ranking; the QS World University Rankings; and the Times Higher Education World University Rankings, or the THE Rankings. The vertical axis tracks the fraction of institutions in our data that found a match in each ranking (total number on top of each bar). The Shanghai ranking has a better coverage, with an entry for ~ 56% of institutional affiliations in our data. Panel (B) shows the fraction of authors who are affiliated to a top 100 institution according to the Shanghai ranking (~ 19.4%).

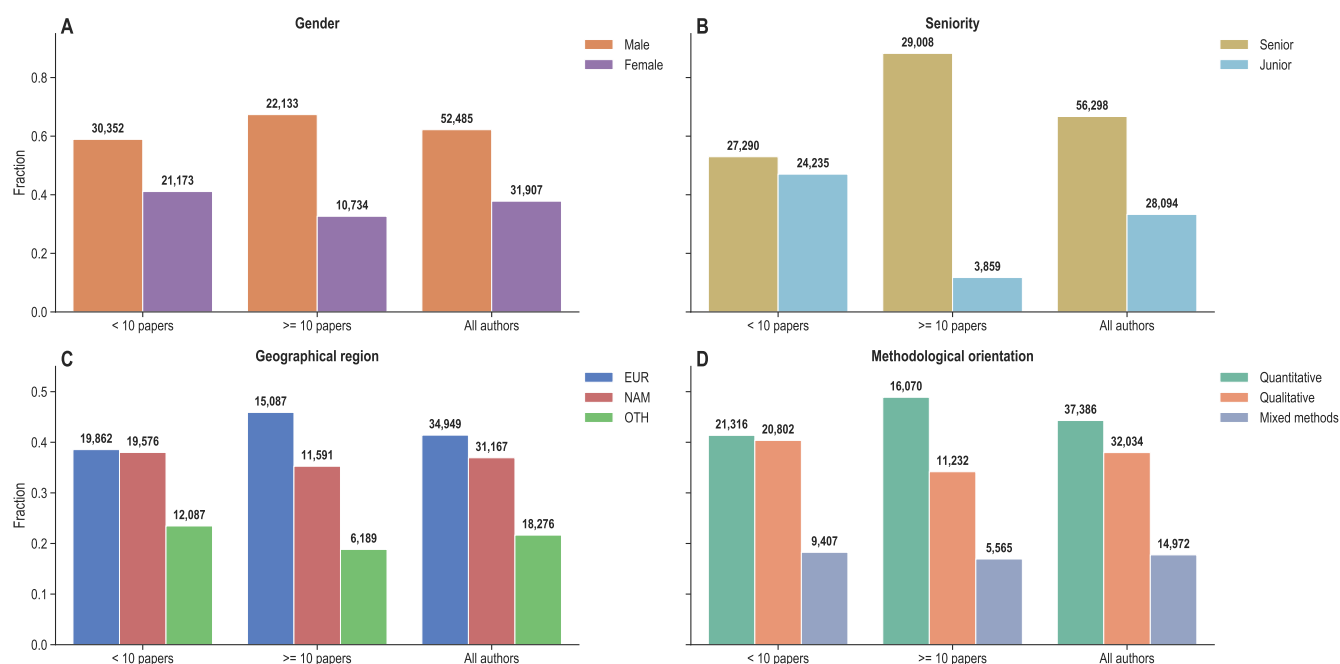


Fig. S14. Impact of the productivity threshold on the data. The four panels show changes in the distribution of co-variables with and without the 10-article threshold. The presence of senior scholars is substantially larger in the filtered dataset (as expected, given the nature of the filter). However, the gender, geographical, and methodological distributions do not change drastically. See also figures [S15](#), [S16](#) and [S17](#) for additional breakdowns.

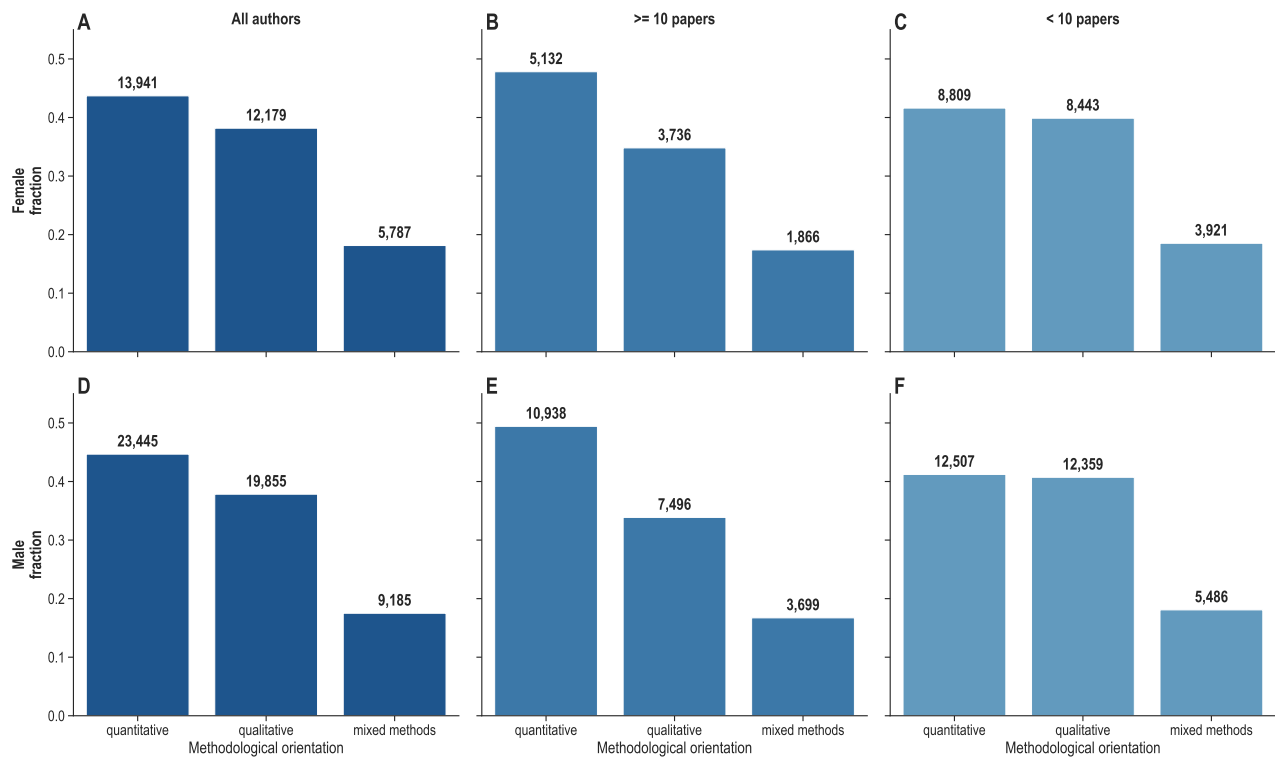


Fig. S15. Gender distribution with and without the productivity threshold. Panels A and D show the total number of female and male scholars in the data, binned according to their main methodological orientation. Panels B and E show the same distribution for scholars that published 10 or more articles (the subset on which our main analyses are based). Panels C and F show the distribution for the subset of scholars with fewer than 10 publications (these are the scholars we remove from our main analyses). On top of each bar, the total count of scholars for that methodological orientation. The results do not indicate that filtering by the total number of papers distorts the gender distribution.

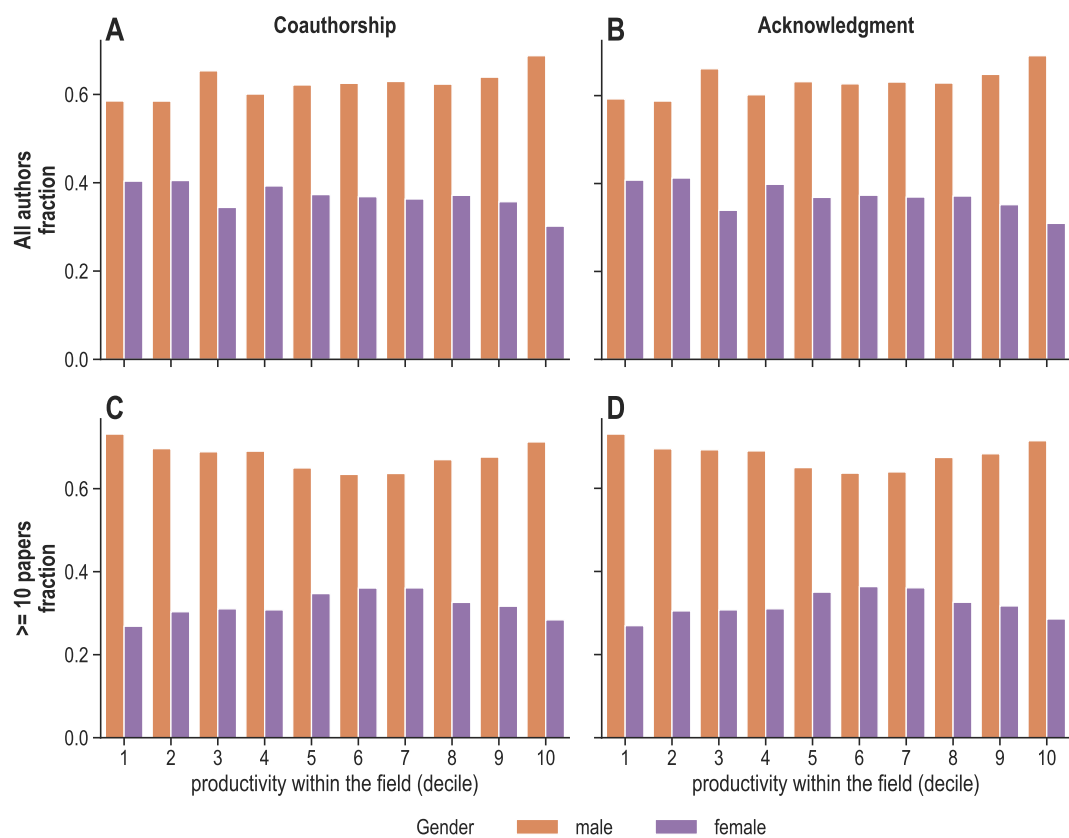


Fig. S16. Scholars in each decile of within-field productivity by gender. The upper row shows the distribution for all scholars in the data, the lower row shows the distribution for scholars in the filtered subset that have 10 or more published articles.

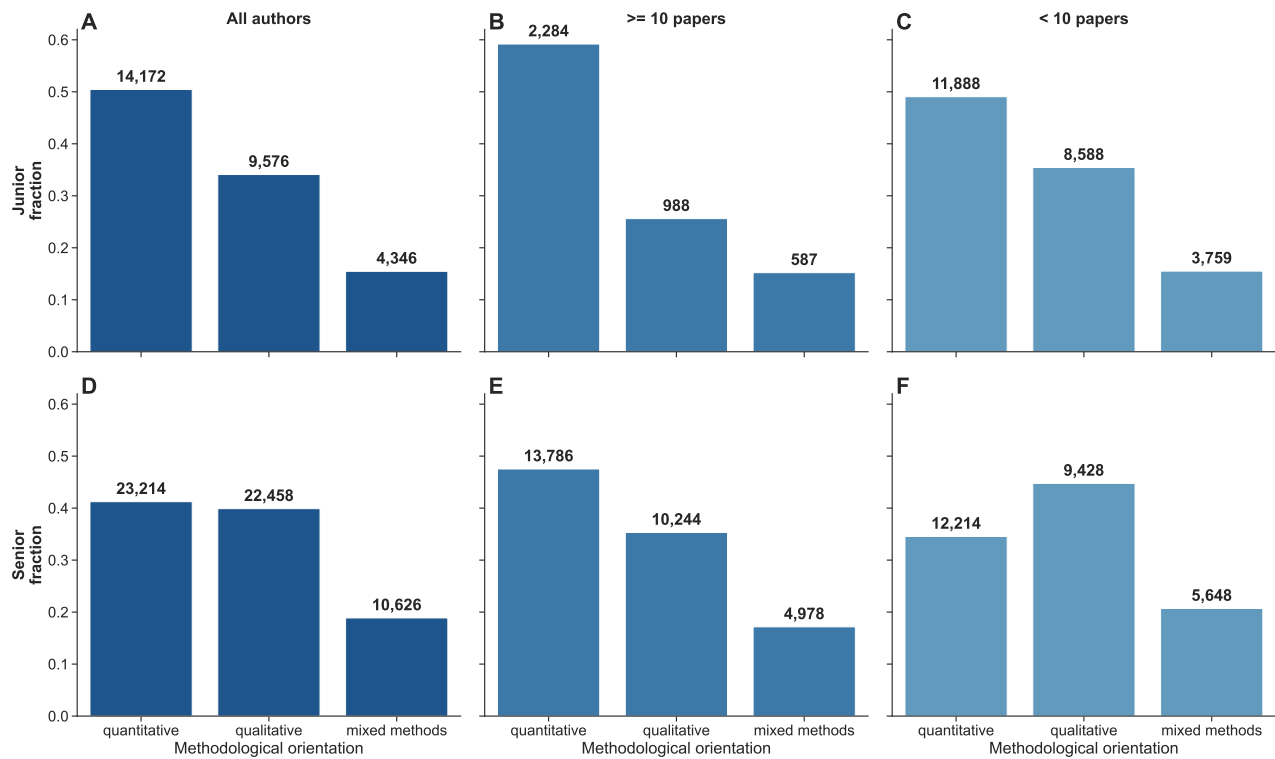


Fig. S17. Seniority distribution with and without the productivity threshold. Panels A and D show the total number of junior and senior scholars in the data, binned according to their main methodological orientation. Panels B and E show the same distribution for scholars that published 10 or more articles (the subset on which our main analyses are based). Panels C and F show the distribution for the subset of scholars with fewer than 10 publications (these are the scholars we remove from our main analyses). On top of each bar, the total count of scholars for that methodological orientation. While the filter removes a larger relative share of junior scholars, it also reduces the number of senior scholars with lower publication counts.

Table S5. Journal composition by method, acknowledgment and impact

Journal	Quant	Formal	Qual/Theor	Ack.	N	SJR	CS
Public Choice	0.608	0.190	0.200	0.830	2018	0.758	3.600
Journal of Politics	0.821	0.056	0.122	0.908	2013	2.792	5.100
Social Science Quarterly	0.771	0.001	0.064	0.537	2011	0.800	3.400
Electoral Studies	0.911	0.000	0.081	0.391	1909	1.796	3.400
PS - Political Science and Politics	0.588	0.015	0.392	0.481	1814	0.919	3.400
Osteuropa	0.000	0.000	0.000	0.000	1777	0.197	0.400
Journal of European Public Policy	0.475	0.009	0.516	0.840	1733	1.967	8.800
Journal of Common Market Studies	0.384	0.011	0.599	0.722	1710	1.127	5.300
Annals of the American Academy of Political and Social Science	0.511	0.004	0.484	0.507	1609	0.961	5.200
Political Geography	0.978	0.000	0.009	0.382	1554	1.371	6.600
Political Quarterly	0.197	0.003	0.555	0.192	1522	0.564	4.200
European Journal of Political Economy	0.944	0.010	0.035	0.551	1472	0.914	3.400
Political Research Quarterly	0.834	0.007	0.141	0.749	1459	1.298	3.600
Survival	0.199	0.001	0.773	0.303	1447	0.349	2.100
Global Policy	0.295	0.010	0.683	0.557	1364	0.697	3.600
Europe-Asia Studies	0.238	0.006	0.496	0.419	1359	0.588	2.600
American Journal of Political Science	0.834	0.081	0.085	0.960	1341	4.629	9.300
International Affairs	0.155	0.001	0.845	0.612	1313	1.322	7.300
Studies in Conflict and Terrorism	0.580	0.015	0.403	0.470	1294	0.331	5.400
West European Politics	0.592	0.005	0.403	0.723	1288	2.395	10.000
International Studies Quarterly	0.688	0.032	0.278	0.885	1273	1.497	4.100
American Political Science Review	0.660	0.077	0.262	0.968	1269	5.070	9.800
Comparative Political Studies	0.789	0.013	0.197	0.904	1267	3.491	8.400
Public Administration	0.504	0.006	0.488	0.645	1261	1.754	7.800
Current History	0.145	0.001	0.834	0.062	1220	0.249	1.600
European Journal of Political Research	0.772	0.010	0.218	0.675	1220	3.334	10.000
Party Politics	0.849	0.012	0.134	0.736	1205	1.878	5.500
Democratization	0.417	0.002	0.582	0.718	1183	1.565	6.400
Journal of Conflict Resolution	0.859	0.085	0.051	0.856	1182	1.860	5.300
Political Studies	0.469	0.008	0.521	0.806	1180	1.793	8.000
Journal of Peace Research	0.839	0.027	0.128	0.917	1158	1.736	6.700
Perspectives on Politics	0.307	0.003	0.687	0.731	1150	1.763	6.000
Journal of Democracy	0.103	0.000	0.599	0.230	1107	1.539	6.300
Political Psychology	0.809	0.003	0.186	0.649	1098	2.161	8.000
British Journal of Political Science	0.836	0.033	0.130	0.910	1093	3.320	8.700
Nationalities Papers	0.180	0.001	0.819	0.580	1082	0.608	3.000
Environmental Politics	0.285	0.009	0.704	0.641	1079	2.375	11.700
Terrorism and Political Violence	0.650	0.010	0.338	0.536	1050	0.659	5.600
Citizenship Studies	0.093	0.001	0.902	0.628	1029	1.059	3.600
Intelligence and National Security	0.224	0.008	0.768	0.547	1012	0.289	1.800
Parliamentary Affairs	0.515	0.003	0.481	0.519	1003	0.576	3.000
Review of International Political Economy	0.247	0.012	0.739	0.812	998	2.256	9.200
Nations and Nationalism	0.218	0.003	0.776	0.606	990	0.731	3.600
International Politics	0.186	0.001	0.812	0.437	974	0.556	3.100
Geopolitics	0.007	0.001	0.088	0.061	958	1.211	7.600
Latin American Perspectives	0.073	0.003	0.915	0.528	953	0.456	1.600
Journal of European Integration	0.057	0.000	0.118	0.131	953	1.181	6.400
American Politics Research	0.970	0.000	0.029	0.688	888	0.944	2.800
Local Government Studies	0.063	0.001	0.044	0.069	873	0.900	4.800
East European Politics and Societies	0.221	0.002	0.775	0.534	873	0.335	1.700
Review of African Political Economy	0.110	0.006	0.754	0.587	866	0.631	3.000
Political Behavior	0.991	0.001	0.008	0.865	862	2.687	8.400
New Political Economy	0.035	0.004	0.112	0.120	851	2.141	10.100
Public Opinion Quarterly	0.970	0.000	0.018	0.876	844	1.636	4.400
Australian Journal of Politics and History	0.178	0.001	0.798	0.406	842	0.138	0.600
British Journal of Politics and International Relations	0.351	0.011	0.637	0.653	818	0.912	4.900
Cambridge Review of International Affairs	0.001	0.000	0.004	0.004	817	0.601	3.600
Historical Materialism	0.033	0.010	0.345	0.141	817	0.422	1.100
Presidential Studies Quarterly	0.453	0.006	0.414	0.499	800	0.716	2.100
Journal of Public Administration Research and Theory	0.829	0.009	0.153	0.696	799	2.981	8.500
Australian Journal of Political Science	0.424	0.006	0.569	0.608	791	0.453	2.100
Armed Forces and Society	0.466	0.009	0.499	0.639	785	0.446	3.000
Politics and Governance	0.414	0.003	0.582	0.860	785	0.703	4.900
Journal of Strategic Studies	0.255	0.005	0.739	0.592	784	0.504	4.000
Political Theory	0.059	0.000	0.940	0.737	778	0.673	2.300
Review of Policy Research	0.538	0.009	0.446	0.534	771	0.699	4.500
Politics and Gender	0.550	0.000	0.448	0.627	765	1.243	6.000
European History Quarterly	0.085	0.005	0.910	0.432	753	0.170	0.600
Governance	0.514	0.007	0.478	0.703	745	1.302	5.700

Continued

Table S5 – continued

Journal	Quant	Formal	Qual/Theor	Ack.	N	SJR	CS
International Feminist Journal of Politics	0.071	0.001	0.912	0.661	731	0.533	3.300
International Political Science Review	0.580	0.007	0.412	0.776	729	0.998	5.600
Policy and Politics	0.000	0.000	0.000	0.000	726	1.111	7.900
Canadian Journal of Political Science	0.550	0.006	0.443	0.746	704	1.186	6.300
International Studies Review	0.244	0.017	0.738	0.591	692	1.241	6.700
Human Rights Quarterly	0.434	0.000	0.307	0.402	691	0.250	1.600
Problems of Post-Communism	0.269	0.004	0.721	0.581	689	0.486	4.000
Policy Studies Journal	0.712	0.013	0.273	0.607	685	1.878	11.600
Socio-Economic Review	0.470	0.012	0.511	0.769	679	1.505	6.800
Forum (Germany)	0.490	0.003	0.477	0.324	679	0.364	1.100
Political Communication	0.791	0.003	0.204	0.608	661	3.348	13.900
New Left Review	0.054	0.000	0.914	0.256	661	0.470	2.200
Communist and Post-Communist Studies	0.986	0.000	0.002	0.429	659	0.407	1.900
Polity	0.184	0.002	0.811	0.716	645	0.274	1.600
International Journal of Public Opinion Research	0.978	0.000	0.017	0.533	643	0.843	3.100
Political Analysis	0.914	0.039	0.047	0.874	641	4.612	8.800
Mediterranean Politics	0.014	0.000	0.071	0.055	634	0.575	4.400
Lex Localis	0.000	0.000	0.000	0.000	625	0.202	1.100
Scottish Journal of Political Economy	0.628	0.191	0.043	0.673	623	0.361	1.800
African Affairs	0.193	0.002	0.804	0.756	623	0.686	4.300
Government and Opposition	0.515	0.003	0.481	0.664	616	1.836	6.700
Regulation and Governance	0.429	0.010	0.560	0.842	602	1.381	7.800
Social Movement Studies	0.022	0.000	0.084	0.087	598	1.421	7.500
International Organization	0.684	0.069	0.247	0.978	594	4.932	14.500
European Union Politics	0.936	0.022	0.042	0.806	593	1.383	4.700
Journal of Contemporary European Studies	0.043	0.002	0.089	0.065	586	0.479	2.700
Telos	0.060	0.002	0.725	0.205	586	0.189	0.200
International Security	0.016	0.000	0.024	0.040	580	1.583	7.400
Politics	0.417	0.003	0.579	0.733	580	0.851	5.500
European Security	0.038	0.002	0.129	0.136	574	0.986	4.100
Contemporary Politics	0.030	0.000	0.051	0.047	570	0.801	4.300
Contemporary Security Policy	0.274	0.004	0.723	0.630	559	1.604	14.600
Political Studies Review	0.423	0.005	0.571	0.582	555	0.815	4.600
International Environmental Agreements: Politics, Law and Economics	0.363	0.058	0.579	0.661	554	1.162	6.000
European Political Science	0.477	0.002	0.506	0.457	551	0.601	3.800
Ethics and International Affairs	0.312	0.000	0.688	0.498	548	0.276	1.900
Comparative European Politics	0.443	0.000	0.557	0.597	548	0.719	3.600
Journal of International Relations and Development	0.185	0.009	0.773	0.770	547	0.690	3.600
Latin American Politics and Society	0.568	0.007	0.425	0.757	544	0.704	3.100
Publius: The Journal of Federalism	0.481	0.007	0.510	0.663	543	0.707	4.000
Cooperation and Conflict	0.193	0.004	0.803	0.671	529	0.756	4.100
South European Society and Politics	0.042	0.000	0.065	0.074	526	0.712	5.800
International Journal of Press/Politics	0.695	0.002	0.303	0.683	505	2.899	12.100
Global Environmental Politics	0.127	0.000	0.770	0.681	504	1.747	7.400
Legislative Studies Quarterly	0.950	0.018	0.022	0.763	502	1.452	2.600
Journal of Political Philosophy	0.068	0.018	0.914	0.928	499	1.609	4.100
Policy and Society	0.000	0.000	0.000	0.000	480	2.221	18.000
Swiss Political Science Review	0.643	0.011	0.332	0.679	473	0.960	3.700
Independent Review	0.000	0.000	0.000	0.000	473	0.179	nan
International Political Sociology	0.138	0.004	0.843	0.689	470	1.111	4.800
Critical Review	0.177	0.004	0.814	0.475	469	0.197	1.300
Journal of Human Rights	0.552	0.000	0.415	0.574	467	0.548	3.100
Politikon	0.000	0.000	0.002	0.000	466	0.278	1.900
State Politics and Policy Quarterly	0.002	0.000	0.002	0.002	465	0.656	2.000
Acta Politica	0.765	0.000	0.225	0.579	463	0.585	2.600
Research and Politics	0.957	0.017	0.022	0.827	463	0.859	2.800
Japanese Journal of Political Science	0.616	0.009	0.376	0.638	458	0.331	1.400
Journal of Elections, Public Opinion and Parties	0.974	0.000	0.026	0.597	457	1.045	4.000
Journal of Information Technology and Politics	0.785	0.002	0.211	0.497	455	1.107	6.600
Irish Political Studies	0.393	0.000	0.607	0.602	450	0.391	2.200
Territory, Politics, Governance	0.238	0.000	0.760	0.800	450	0.789	5.600
German Politics	0.482	0.007	0.511	0.531	448	0.561	3.800
Politics and Religion	0.604	0.000	0.396	0.583	444	0.494	3.000
Political Science Quarterly	0.360	0.002	0.635	0.464	444	0.268	1.700
Journal of Theoretical Politics	0.300	0.572	0.086	0.836	444	0.370	2.100
British Politics	0.342	0.009	0.649	0.453	444	0.352	2.400
Journal of Policy History	0.190	0.000	0.810	0.657	443	0.132	0.500
Studies in Comparative International Development	0.477	0.005	0.518	0.749	442	0.967	4.400
Contemporary Political Theory	0.057	0.000	0.918	0.556	441	0.630	3.300
Politics and Society	0.202	0.005	0.789	0.939	440	0.885	5.000

Continued

Table S5 – continued

Journal	Quant	Formal	Qual/Theor	Ack.	N	SJR	CS
Contemporary Southeast Asia	0.130	0.000	0.328	0.287	439	0.392	2.000
Critical Policy Studies	0.011	0.002	0.073	0.066	437	0.544	3.500
Post-Soviet Affairs	0.528	0.007	0.447	0.709	436	1.170	5.000
Comparative Politics	0.363	0.007	0.356	0.602	430	0.815	4.600
Journal of Public Policy	0.640	0.057	0.303	0.806	422	0.873	4.400
Scandinavian Political Studies	0.606	0.010	0.257	0.469	416	0.843	2.400
Review of International Organizations	0.705	0.045	0.251	0.797	403	1.564	8.100
Journal of Chinese Political Science	0.420	0.010	0.570	0.532	395	1.082	6.800
Politics, Philosophy and Economics	0.076	0.064	0.858	0.898	393	0.745	1.600
Journal of Political Ideologies	0.015	0.000	0.077	0.056	392	0.343	3.300
Economics and Politics	0.714	0.227	0.057	0.849	384	0.662	2.400
Business and Politics	0.596	0.031	0.373	0.617	381	0.648	3.600
European Political Science Review	0.743	0.013	0.244	0.729	377	1.710	5.700
Journal of Women, Politics and Policy	0.607	0.000	0.393	0.455	374	0.559	1.800
World Politics	0.563	0.021	0.397	0.791	373	3.025	8.400
East European Politics	0.399	0.005	0.582	0.647	368	1.262	7.000
Political Science Research and Methods	0.938	0.028	0.034	0.826	356	2.431	8.100
The International Journal of Transitional Justice	0.085	0.003	0.912	0.686	331	0.346	3.300
Politicka Ekonomie	0.580	0.123	0.074	0.193	326	0.133	0.500
Journal of Cold War Studies	0.000	0.000	0.003	0.003	318	0.249	1.200
Policy and Internet	0.630	0.000	0.367	0.433	300	1.977	8.400
Philosophy and Public Affairs	0.070	0.022	0.905	0.941	273	1.596	3.900
Journal of Australian Political Economy	0.000	0.000	0.000	0.000	272	0.411	3.200
Political Science	0.504	0.016	0.477	0.547	258	0.175	0.900
International Theory	0.070	0.051	0.875	0.856	257	1.342	6.600
Quarterly Journal of Political Science	0.702	0.282	0.008	0.899	248	2.033	3.500
Studies in American Political Development	0.172	0.000	0.828	0.765	238	1.032	1.300
Ethics and Global Politics	0.080	0.005	0.915	0.667	213	0.171	1.100
Revista Brasileira de Política Internacional	0.219	0.005	0.757	0.343	210	0.349	1.800
Journal of Chinese Governance	0.015	0.000	0.034	0.024	205	0.796	7.400
International Journal of Conflict and Violence	0.561	0.000	0.439	0.490	198	0.461	5.200
Politix	0.000	0.000	0.000	0.000	184	0.124	0.700
Journal of Current Southeast Asian Affairs	0.231	0.000	0.769	0.588	160	0.780	5.000
Peacebuilding	0.046	0.000	0.250	0.178	152	0.304	2.700
Romanian Journal of Political Science	0.000	0.000	0.000	0.000	109	0.105	0.300

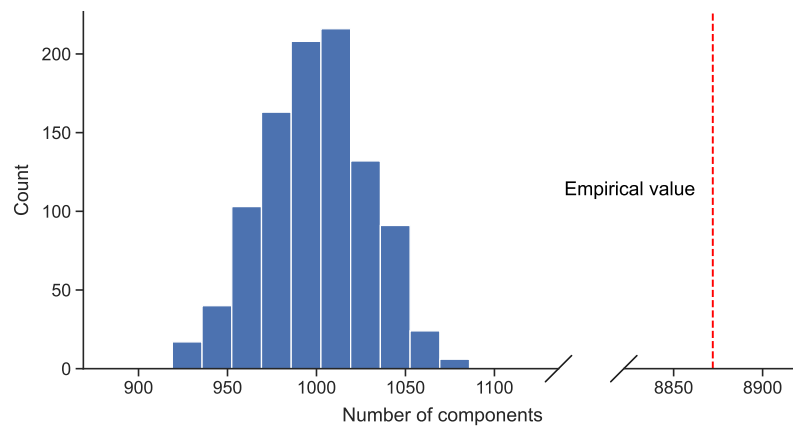


Fig. S18. Connectivity in co-authorship network. The dashed red line marks the observed number of components, the histogram plots the distribution in the number of components for 1,000 networks randomly assembled according to the configuration model (2).

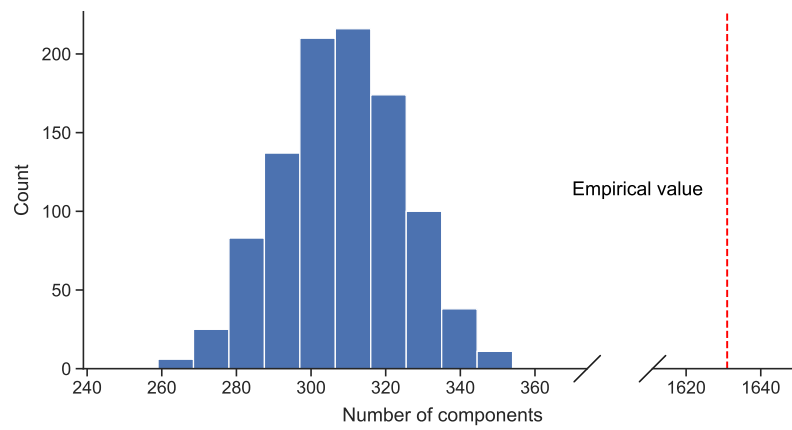


Fig. S19. Connectivity in acknowledgment network. The dashed red line marks the observed number of components, the histogram plots the distribution in the number of components for 1,000 networks randomly assembled according to the configuration model (2).

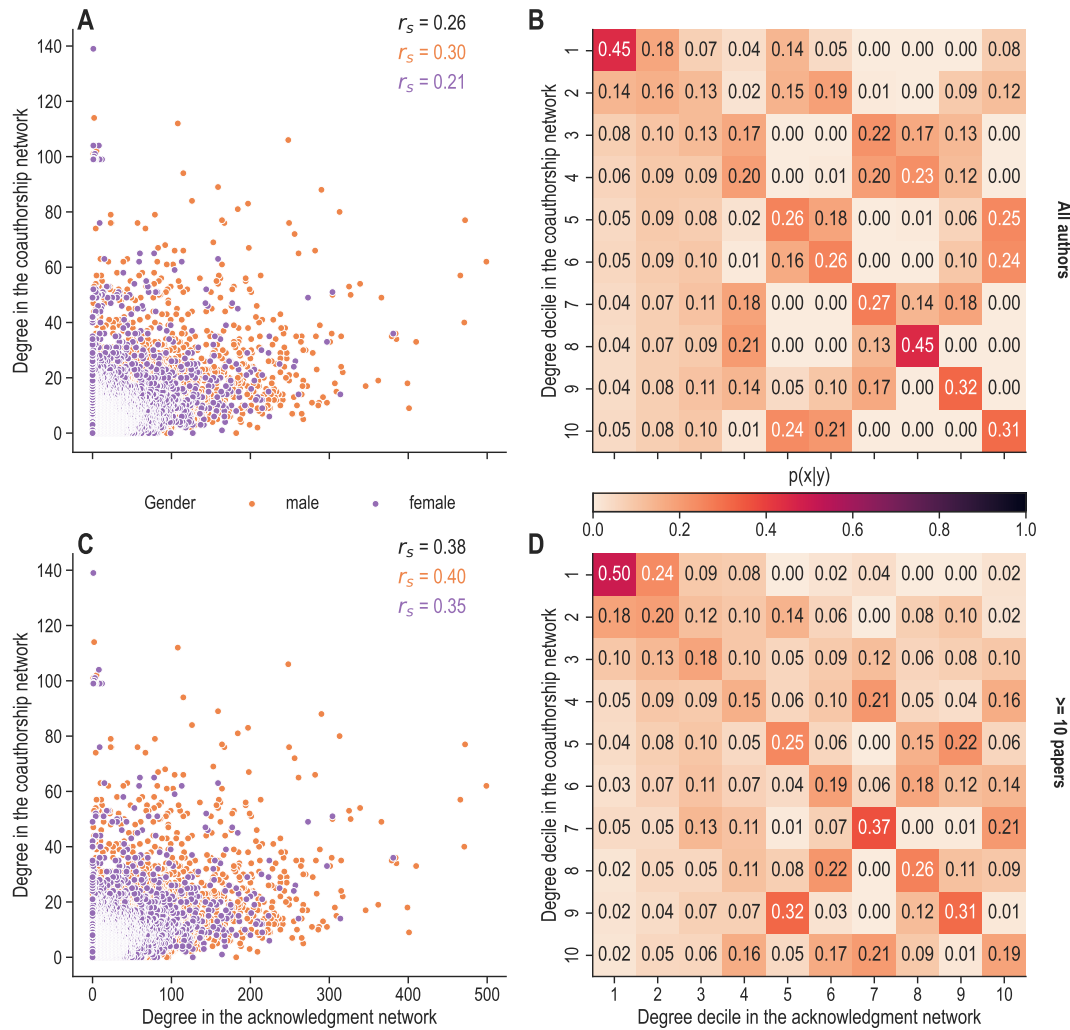


Fig. S20. Overlap of scholars in the two networks. The upper row shows the correlation of the degree centrality of all scholars present in the co-authorship and acknowledgment networks. The lower row shows the same correlation for the subset of the data that only retains scholars with 10 publications or more. Panels A and C display the positions of each author in the degree distribution for both networks. Annotations indicate the spearman rank correlation coefficient for all, male and female scholars, respectively. Panels B and D display the correlation coefficients for every decile of the degree distribution. See table S6 for additional summary statistics. The correlation between centrality in both networks is moderate to weak, indicating that these structures capture fundamentally different contexts.

Table S6. Characteristics of the Two Networks by Gender (Overlapping Scholars)

Statistics	Co-Authorship	Acknowledgment
Male	52,485 (62.71%)	52, 485 (62.71%)
Female	31,204 (37.28%)	31,204 (37.28%)
Degree range (male)	[0, 114]	[0, 499]
Degree range (female)	[0, 139]	[0, 381]
Average degree (male)	2.82	8.31
Average degree (female)	2.93	5.77
Median degree (male)	1.0	1.0
Median degree (female)	1.0	1.0

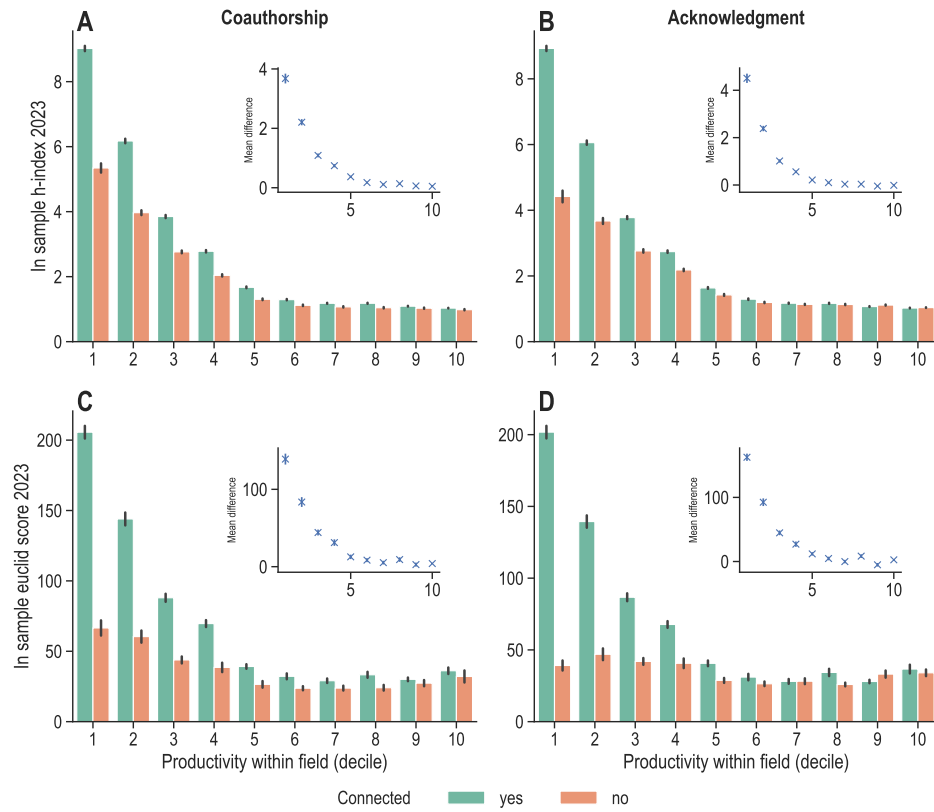


Fig. S21. Differences in publication impact. Extension of Figure 2 in the main text using the networks as they existed in 2018. We observe a similar trend but the difference between the effect in the acknowledgment and collaboration networks is reduced.

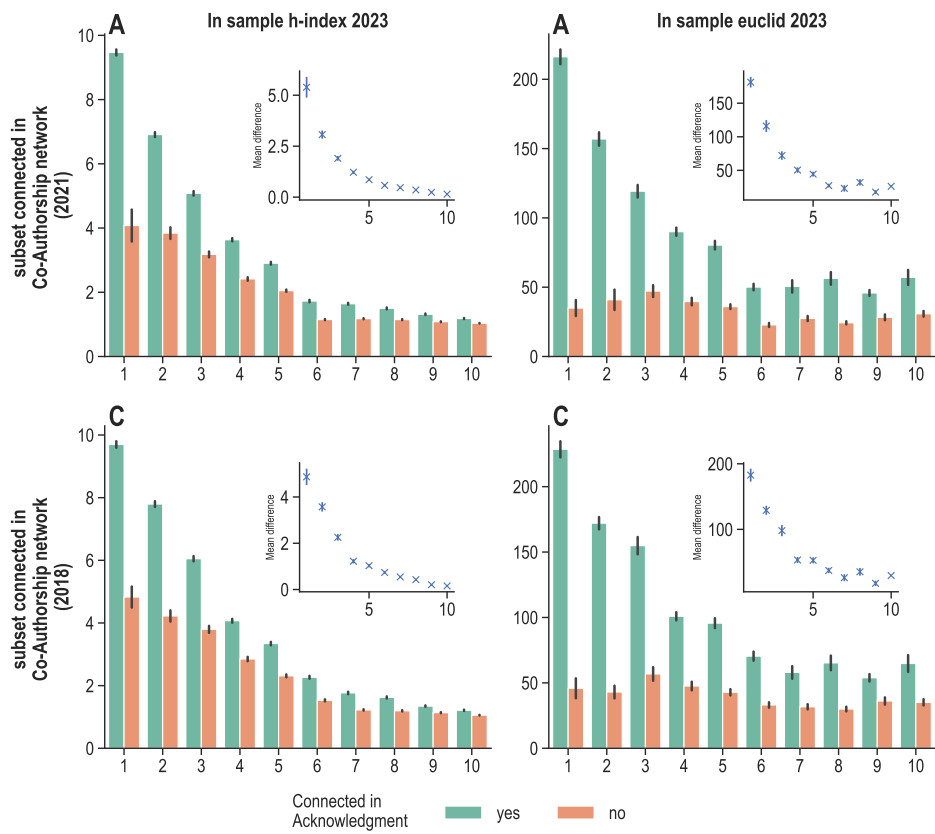


Fig. S22. Differences in publication impact. Extension of Figure 2 in the main text and to [S11](#) using the subset of scholars connected in the co-authorship network to measure group differences. Upper row corresponds to the 2021 network data and lower row to the 2018 network data.

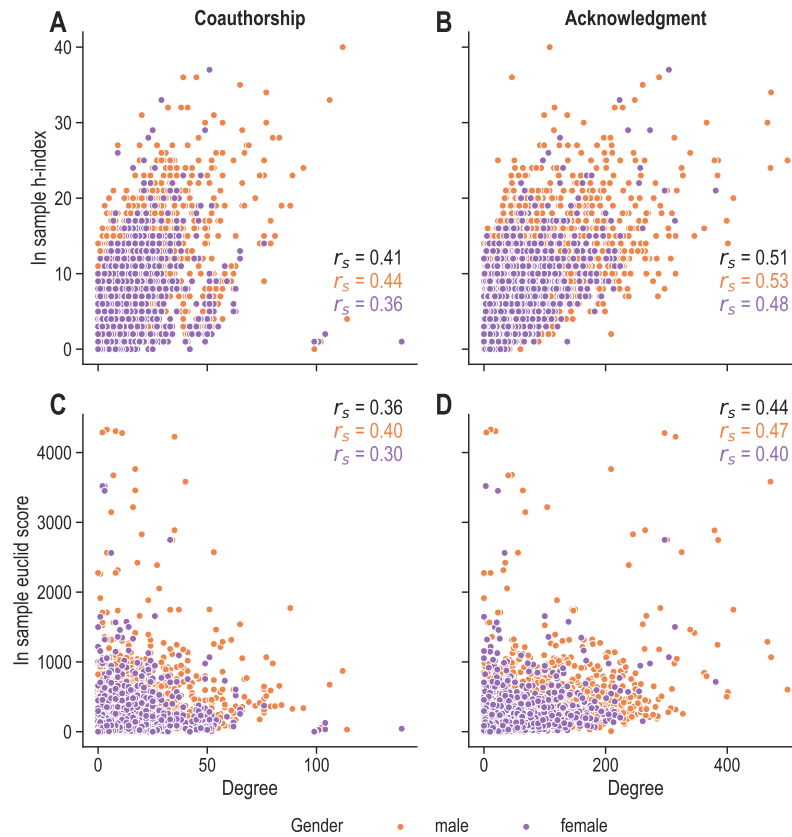


Fig. S23. Correlation of degree with publication impact. The upper row shows the correlation between degree centrality in both networks (measured in 2021) and publication impact, as measured by the h-index (2023). The lower row shows the correlation between degree centrality and the Euclid scores (also measured in 2021 and 2023, respectively). Annotations indicate the spearman rank correlation coefficient for all, male, and female scholars. The association of centrality and impact is slightly weaker for the female scholars.

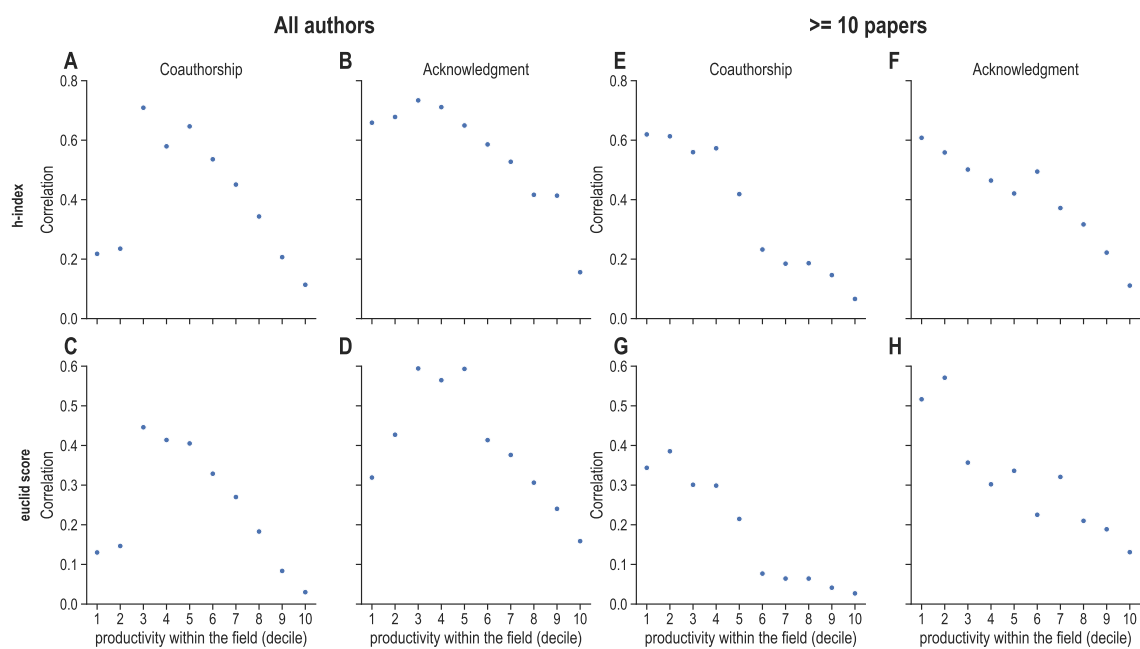


Fig. S24. Correlation of degree with publication impact by decile in productivity within the field. The upper row shows the correlations between degree centrality and the h-index, before and after applying the 10-paper filter. The lower row shows the correlation between degree centrality and the Euclid scores, before and after the filter. The smoothing effect of the 10-paper filter clarifies the signal in the upper productivity deciles, where scholars with only a few publications (but all of them in Political Science) introduce noisy observations.

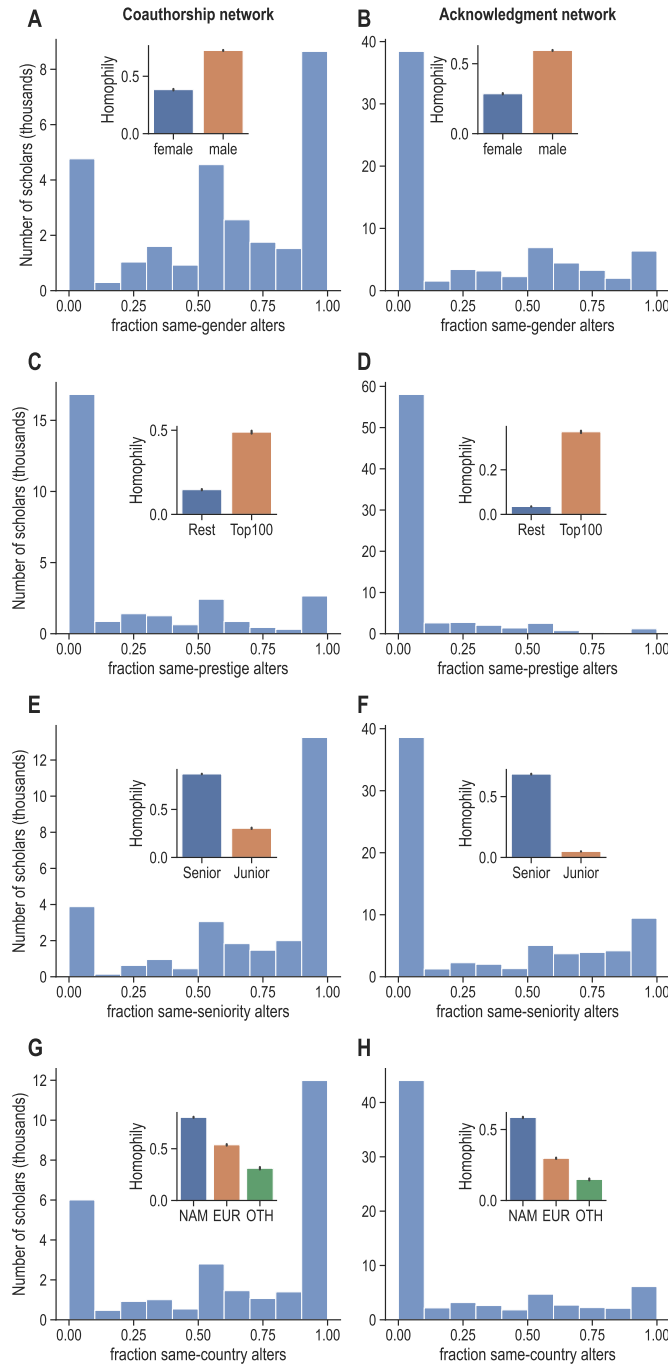


Fig. S25. Fraction of connections to similar others. The histograms show the fraction of ties that scholars in the two networks have with similar others, across four dimensions: (A-B) gender, (C-D) institutional prestige, (E-F) seniority, and (G-H) country of affiliation. Insets show the average fraction within each group category. Isolated nodes are excluded from this plot. These distributions reveal different behavioral patterns depending on the network. For instance, there is less geographical homophily in the acknowledgment network than in the co-authorship network, likely due to the higher organizational cost of the latter compared to the former. Likewise, there are substantially more junior-senior connections in the acknowledgment network.

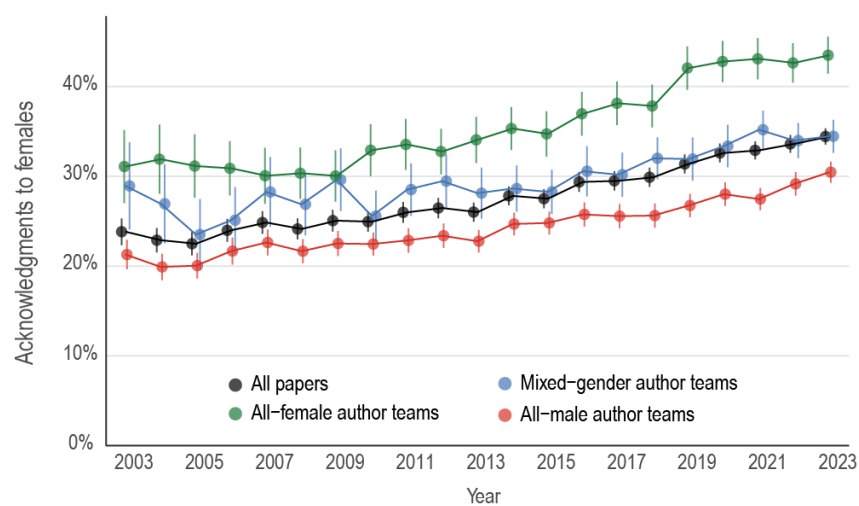


Fig. S26. Fraction of articles acknowledging females. The number of papers that acknowledge female scholars has increased more than 10 percentage points over the past twenty years. The largest fraction of these papers are written by all-female author teams.

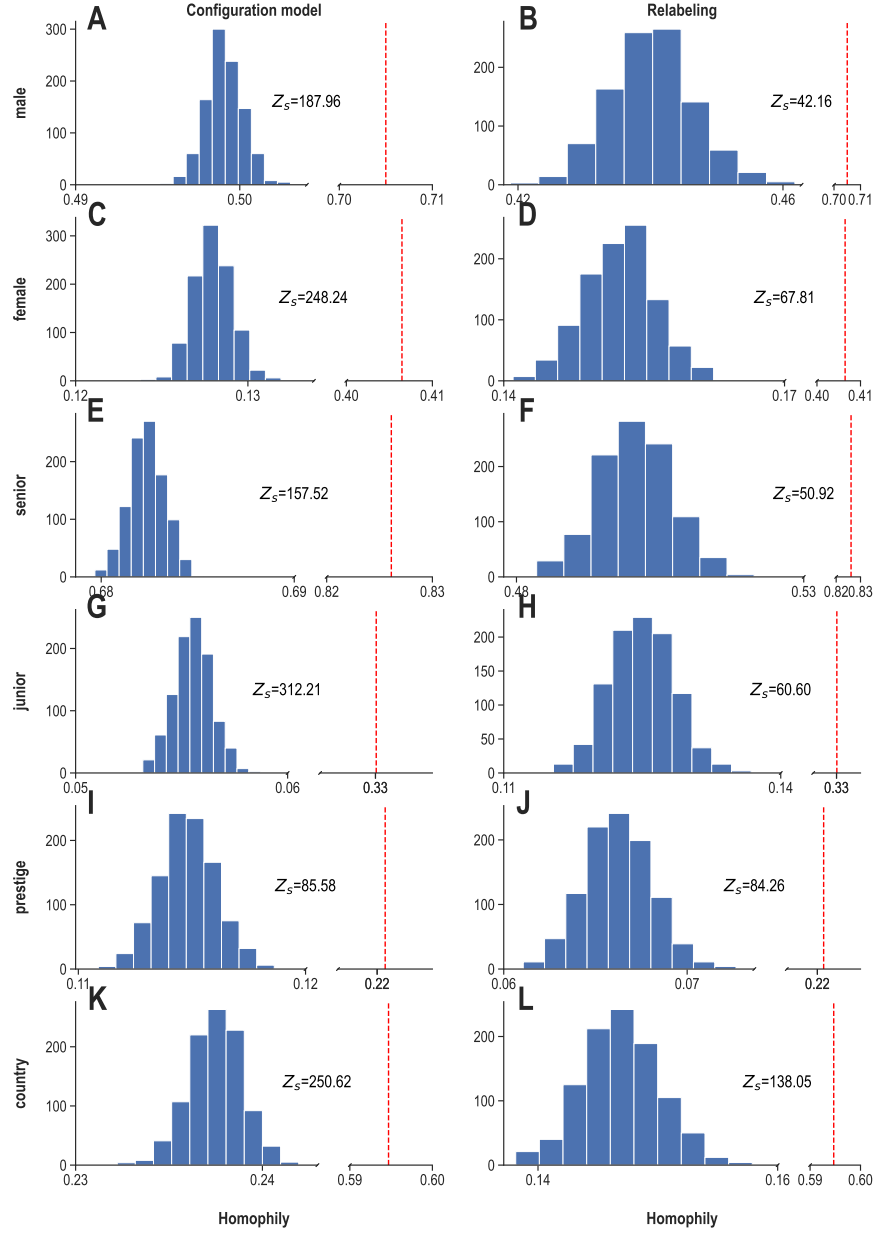


Fig. S27. Compositional and behavioral homophily for the co-authorship network. The histograms show the distribution of values returned by 1,000 random permutations of the network ties (left column) and 1,000 random permutations of the node labels (right column). The dashed vertical line indicates the degree-weighted average homophily in the observed network. Z_s denotes the normalized (z-score) distance between the random mean and the observed value.

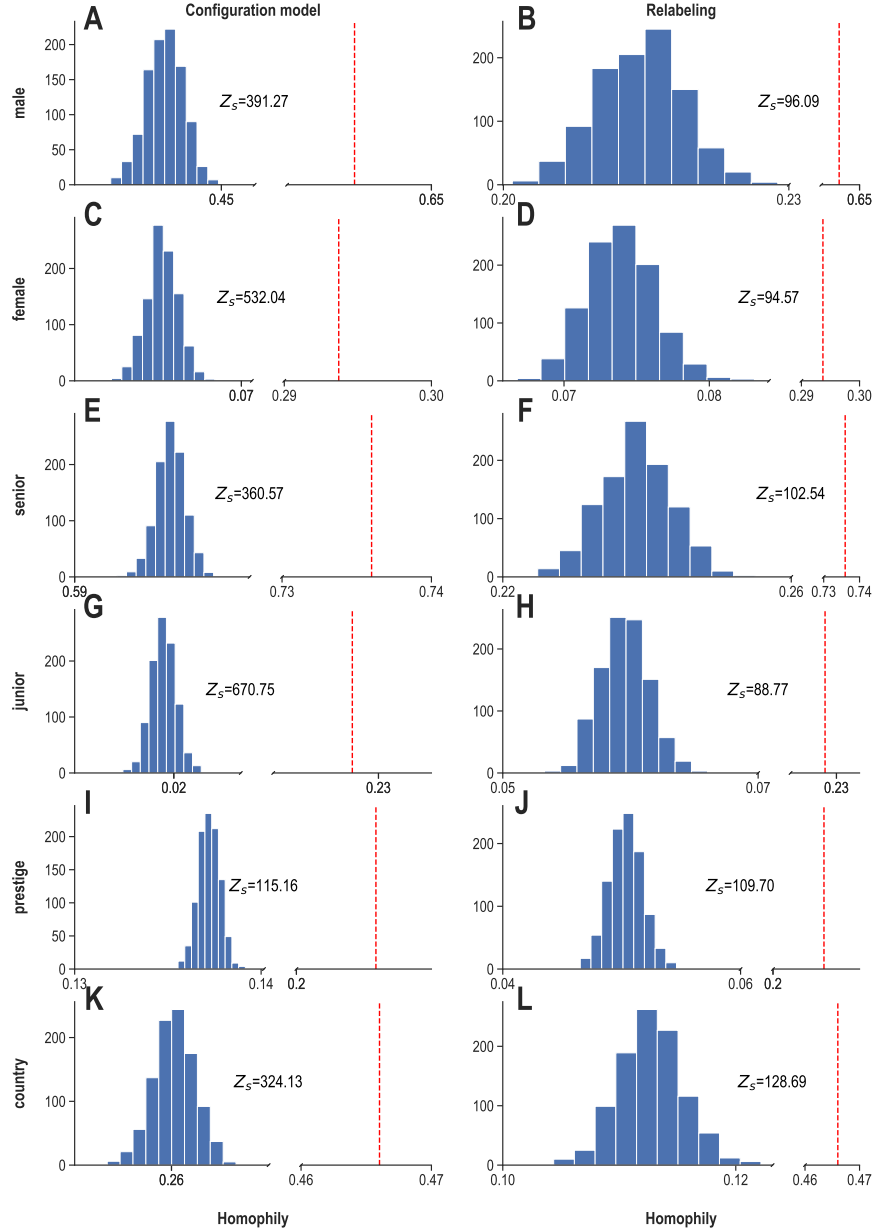


Fig. S28. Compositional and behavioral homophily for the acknowledgment network. The histograms show the distribution of values returned by 1,000 random permutations of the network ties (left column) and 1,000 random permutations of the node labels (right column). The dashed vertical line indicates the degree-weighted average homophily in the observed network. Z_s denotes the normalized (z-score) distance between the random mean and the observed value.

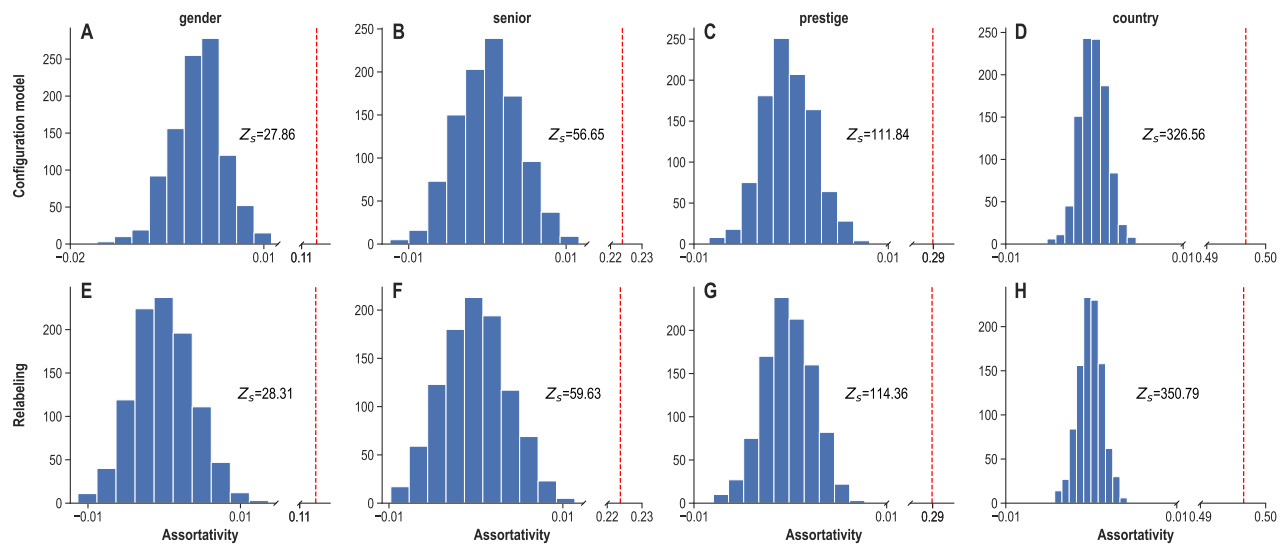


Fig. S29. Compositional and behavioral assortativity for the co-authorship network. The histograms show the distribution of values returned by 1,000 random permutations of the network ties (A-F, upper row) and 1,000 random permutations of the node labels (G-L, lower row). Dashed vertical lines indicate the magnitude of the observed assortativity in the co-authorship network. Z_s denotes the normalized (z-score) distance between the random mean and the observed value.

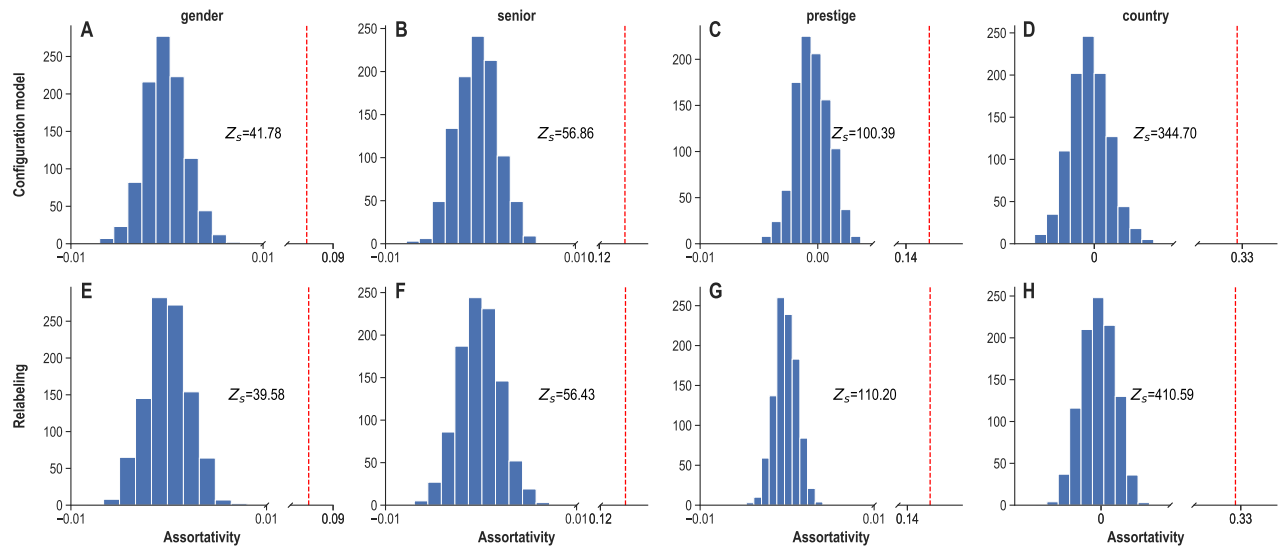


Fig. S30. Compositional and behavioral assortativity for the acknowledgment network. The histograms show the distribution of values returned by 1,000 random permutations of the network ties (A-F, upper row) and 1,000 random permutations of the node labels (G-L, lower row). Dashed vertical lines indicate the magnitude of the observed assortativity in the acknowledgment network. Z_s denotes the normalized (z-score) distance between the random mean and the observed value.

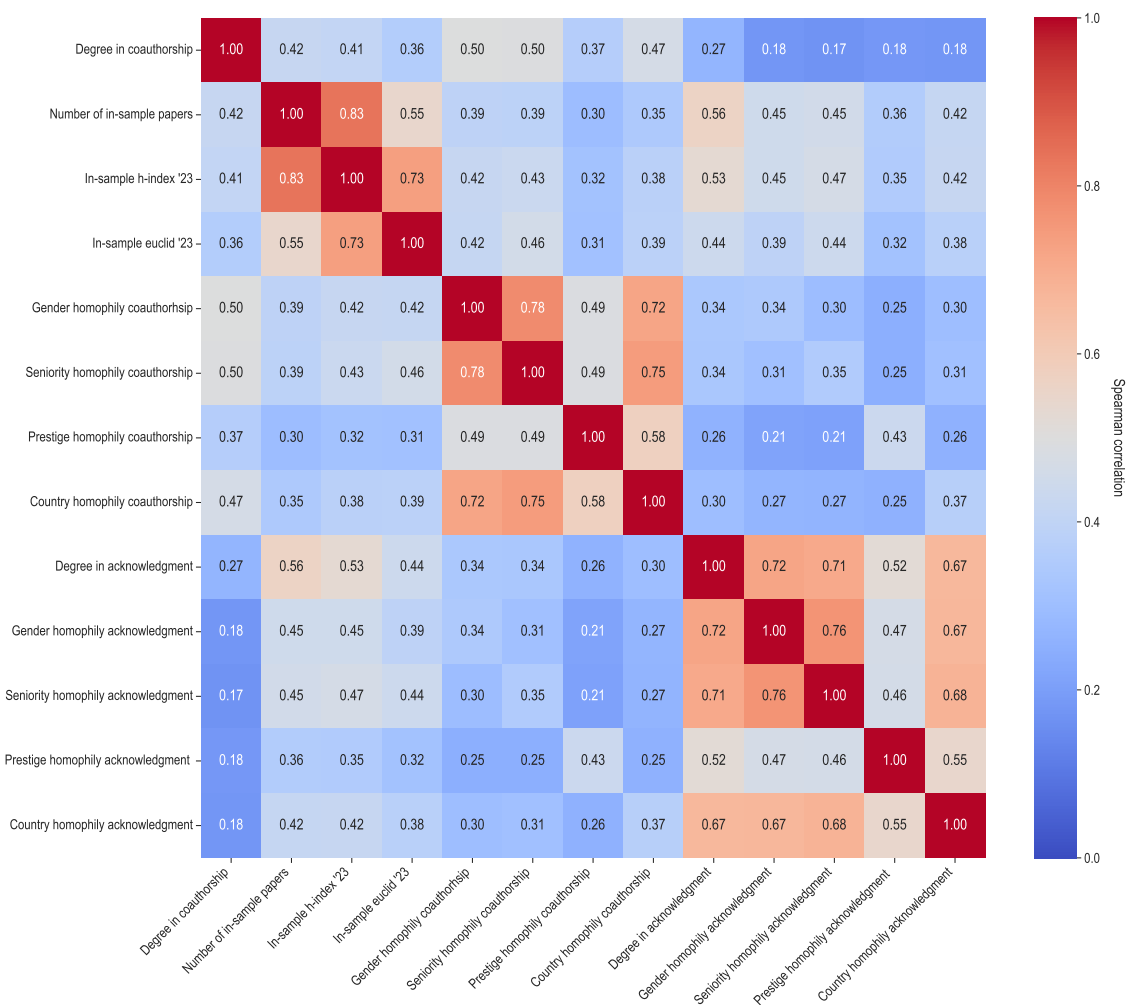


Fig. S31. Correlation matrix. Spearman rank correlation coefficients between all non-binary variables considered in the regression and classifier models. Warmer colors indicate higher correlation.

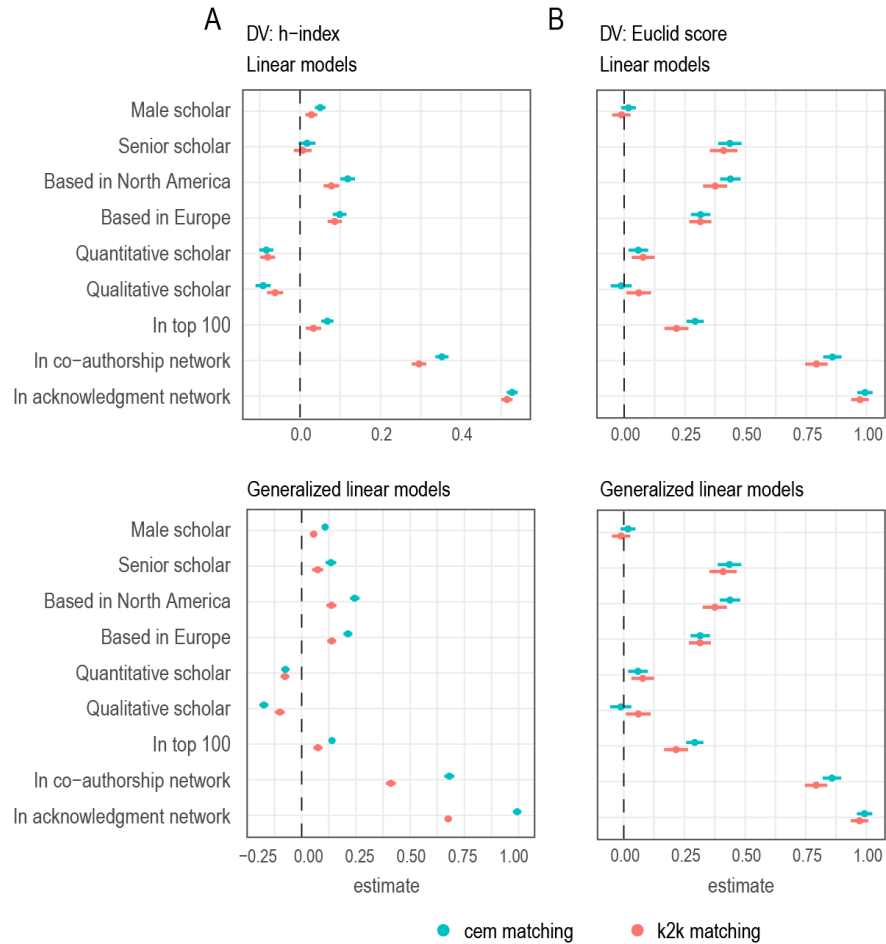


Fig. S32. Regression estimates for matched data. Extension to Figure 3A-B in the main text. The k -to- k matching is more stringent than the unbalanced CEM, which results in more unmatched cases. This plot shows that when using the unbalanced stratified matching, results are also consistent with those reported in the main text. See also Tables S7-S10 for full regression outputs.

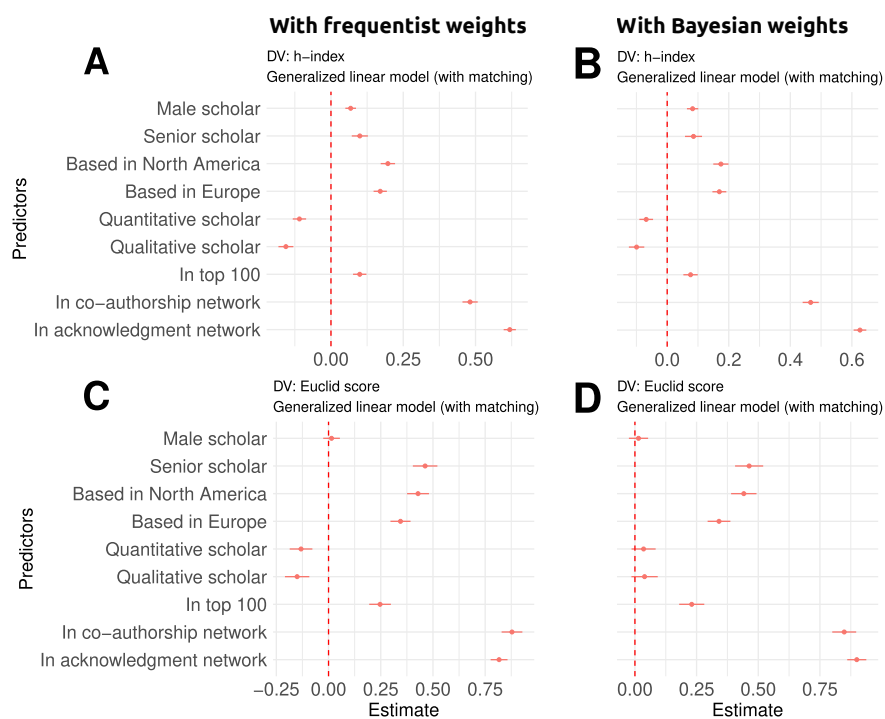


Fig. S33. Regression estimates for weighted matched data. Extension to Figure 3A-B in the main text. Here we show the regression results using the frequentist weights (left column) and the Bayesian weights (right column). Results are consistent when using the h-index as the DV: informal ties are the most relevant predictor. When using the Euclid Score as the DV, informal connections are still a strong and significant predictor of impact, but the magnitude is not statistically distinguishable from formal connections. See also Tables S11-S18 for full regression outputs. (See section 4 for details on weight calculation).

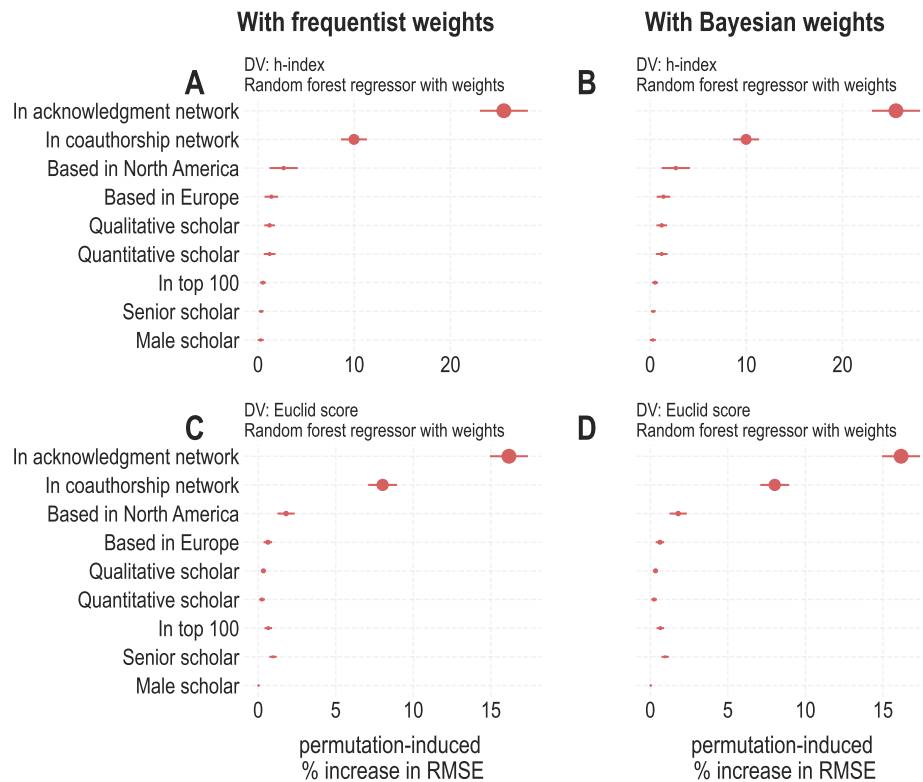


Fig. S34. Random Forests regression estimates using weighted data. Extension to Figure 3C-D in the main text. Informal ties remain the most relevant predictor under the frequentist (A,C) and Bayesian (B,D) weighting schemes. (See section 4 for details on weight calculation).

Table S7: Linear models | DV: h-index

	(1) k2k matching	(1) cem matching	(2) k2k matching	(2) cem matching	(3) k2k matching	(3) cem matching
Intercept	1.040*** (0.018)	0.878*** (0.016)	0.786*** (0.019)	0.559*** (0.017)	0.529*** (0.017)	0.449*** (0.016)
Male scholar	0.017* (0.009)	0.080*** (0.008)	0.028*** (0.008)	0.086*** (0.007)	0.028*** (0.007)	0.050*** (0.007)
Senior scholar	−0.004 (0.013)	0.054*** (0.012)	0.007 (0.013)	0.060*** (0.011)	0.007 (0.011)	0.017+ (0.010)
Based in North America	0.064*** (0.011)	0.206*** (0.010)	0.078*** (0.011)	0.206*** (0.010)	0.078*** (0.010)	0.118*** (0.009)
Based in Europe	0.092*** (0.010)	0.169*** (0.010)	0.087*** (0.010)	0.156*** (0.009)	0.086*** (0.009)	0.098*** (0.009)
Quantitative scholar	−0.053*** (0.011)	0.043*** (0.010)	−0.080*** (0.011)	−0.004 (0.010)	−0.080*** (0.009)	−0.084*** (0.009)
Qualitative scholar	−0.108*** (0.012)	−0.031** (0.010)	−0.062*** (0.011)	0.016 (0.010)	−0.062*** (0.010)	−0.092*** (0.009)
In top 100	0.025* (0.011)	0.122*** (0.009)	0.033** (0.011)	0.121*** (0.008)	0.033*** (0.010)	0.068*** (0.008)
In co-authorship network			0.296*** (0.010)	0.399*** (0.009)	0.296*** (0.009)	0.353*** (0.008)
In acknowledgment network					0.514*** (0.007)	0.527*** (0.007)
Num.Obs.	18 269	29 950	18 269	29 950	18 269	29 864
R2	0.010	0.031	0.053	0.091	0.267	0.241
R2 Adj.	0.009	0.031	0.052	0.091	0.267	0.241

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table S8: Generalized linear models | DV: h-index

	(1) k2k matching	(1) cem matching	(2) k2k matching	(2) cem matching	(3) k2k matching	(3) cem matching
Intercept	0.718*** (0.021)	0.456*** (0.017)	0.169*** (0.025)	-0.187*** (0.020)	0.360*** (0.021)	-0.576*** (0.021)
Male scholar	0.059*** (0.010)	0.155*** (0.007)	0.078*** (0.010)	0.164*** (0.007)	0.055*** (0.008)	0.108*** (0.007)
Senior scholar	0.086*** (0.015)	0.193*** (0.012)	0.104*** (0.015)	0.201*** (0.012)	0.073*** (0.013)	0.134*** (0.012)
Based in North America	0.171*** (0.014)	0.382*** (0.011)	0.196*** (0.014)	0.382*** (0.011)	0.137*** (0.011)	0.244*** (0.011)
Based in Europe	0.207*** (0.013)	0.323*** (0.010)	0.197*** (0.013)	0.303*** (0.010)	0.138*** (0.010)	0.213*** (0.010)
Quantitative scholar	-0.061*** (0.012)	0.101*** (0.009)	-0.108*** (0.012)	0.034*** (0.009)	-0.078*** (0.010)	-0.075*** (0.009)
Qualitative scholar	-0.225*** (0.013)	-0.104*** (0.010)	-0.142*** (0.014)	-0.028** (0.010)	-0.101*** (0.011)	-0.175*** (0.010)
In top 100	0.090*** (0.013)	0.219*** (0.008)	0.105*** (0.013)	0.218*** (0.008)	0.075*** (0.011)	0.139*** (0.008)
In co-authorship network			0.616*** (0.015)	0.756*** (0.011)	0.411*** (0.012)	0.680*** (0.011)
In acknowledgment network					0.675*** (0.008)	0.993*** (0.010)
Num.Obs.	18 269	29 950	18 269	29 950	18 269	29 864
AIC	83 184.4	153 773.7	81 171.0	148 552.4	78 407.6	135 257.9

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table S9: Linear models | DV: Euclid score

	(1) k2k matching	(1) cem matching	(2) k2k matching	(2) cem matching	(3) k2k matching	(3) cem matching
Intercept	2.430*** (0.043)	2.135*** (0.036)	1.747*** (0.047)	1.380*** (0.039)	1.261*** (0.045)	1.167*** (0.037)
Male scholar	-0.042* (0.021)	0.068*** (0.017)	-0.011 (0.021)	0.082*** (0.017)	-0.011 (0.019)	0.018 (0.016)
Senior scholar	0.380*** (0.032)	0.502*** (0.027)	0.410*** (0.031)	0.518*** (0.026)	0.410*** (0.029)	0.436*** (0.025)
Based in North America	0.338*** (0.028)	0.596*** (0.023)	0.375*** (0.027)	0.597*** (0.022)	0.375*** (0.025)	0.438*** (0.021)
Based in Europe	0.329*** (0.025)	0.449*** (0.022)	0.314*** (0.025)	0.418*** (0.021)	0.314*** (0.023)	0.315*** (0.020)
Quantitative scholar	0.151*** (0.026)	0.323*** (0.022)	0.079** (0.026)	0.213*** (0.022)	0.078** (0.024)	0.059** (0.021)
Qualitative scholar	-0.064* (0.028)	0.085*** (0.024)	0.060* (0.028)	0.196*** (0.023)	0.060* (0.026)	-0.012 (0.022)
In top 100	0.194*** (0.028)	0.396*** (0.020)	0.216*** (0.027)	0.394*** (0.019)	0.216*** (0.025)	0.292*** (0.018)
In co-authorship network			0.793*** (0.025)	0.944*** (0.020)	0.793*** (0.023)	0.859*** (0.019)
In acknowledgment network					0.972*** (0.018)	0.993*** (0.016)
Num.Obs.	18 264	29 941	18 264	29 941	18 264	29 855
R2	0.026	0.062	0.077	0.125	0.202	0.225
R2 Adj.	0.026	0.062	0.077	0.125	0.202	0.225

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table S10: Generalized linear models | DV: Euclid score

	(1) k2k matching	(1) cem matching	(2) k2k matching	(2) cem matching	(3) k2k matching	(3) cem matching
Intercept	2.430*** (0.043)	2.135*** (0.036)	1.747*** (0.047)	1.380*** (0.039)	1.261*** (0.045)	1.167*** (0.037)
Male scholar	−0.042* (0.021)	0.068*** (0.017)	−0.011 (0.021)	0.082*** (0.017)	−0.011 (0.019)	0.018 (0.016)
Senior scholar	0.380*** (0.032)	0.502*** (0.027)	0.410*** (0.031)	0.518*** (0.026)	0.410*** (0.029)	0.436*** (0.025)
Based in North America	0.338*** (0.028)	0.596*** (0.023)	0.375*** (0.027)	0.597*** (0.022)	0.375*** (0.025)	0.438*** (0.021)
Based in Europe	0.329*** (0.025)	0.449*** (0.022)	0.314*** (0.025)	0.418*** (0.021)	0.314*** (0.023)	0.315*** (0.020)
Quantitative scholar	0.151*** (0.026)	0.323*** (0.022)	0.079** (0.026)	0.213*** (0.022)	0.078** (0.024)	0.059** (0.021)
Qualitative scholar	−0.064* (0.028)	0.085*** (0.024)	0.060* (0.028)	0.196*** (0.023)	0.060* (0.026)	−0.012 (0.022)
In top 100	0.194*** (0.028)	0.396*** (0.020)	0.216*** (0.027)	0.394*** (0.019)	0.216*** (0.025)	0.292*** (0.018)
In co-authorship network			0.793*** (0.025)	0.944*** (0.020)	0.793*** (0.023)	0.859*** (0.019)
In acknowledgment network					0.972*** (0.018)	0.993*** (0.016)
Num.Obs.	18 264	29 941	18 264	29 941	18 264	29 855
AIC	62 930.8	104 344.1	61 957.6	102 276.4	59 293.9	98 292.2

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table S11: Weighted linear models (frequentist) | DV: h-index

	(1) k2k matching	(1) cem matching	(2) k2k matching	(2) cem matching	(3) k2k matching	(3) cem matching
Intercept	1.254*** (0.020)	1.146*** (0.018)	0.954*** (0.022)	0.784*** (0.019)	0.560*** (0.022)	0.489*** (0.020)
Male scholar	0.024* (0.010)	0.070*** (0.008)	0.036*** (0.009)	0.076*** (0.008)	0.039*** (0.009)	0.059*** (0.008)
Senior scholar	0.015 (0.014)	0.055*** (0.012)	0.026+ (0.014)	0.059*** (0.012)	0.016 (0.013)	0.017 (0.012)
Based in North America	0.116*** (0.013)	0.193*** (0.011)	0.130*** (0.012)	0.193*** (0.010)	0.133*** (0.011)	0.147*** (0.010)
Based in Europe	0.127*** (0.012)	0.180*** (0.010)	0.120*** (0.011)	0.166*** (0.010)	0.116*** (0.010)	0.128*** (0.010)
Quantitative scholar	-0.221*** (0.013)	-0.149*** (0.011)	-0.253*** (0.013)	-0.178*** (0.011)	-0.158*** (0.012)	-0.144*** (0.011)
Qualitative scholar	-0.295*** (0.014)	-0.240*** (0.012)	-0.243*** (0.014)	-0.171*** (0.012)	-0.141*** (0.013)	-0.153*** (0.011)
In top 100	0.046*** (0.013)	0.106*** (0.009)	0.053*** (0.012)	0.107*** (0.008)	0.053*** (0.011)	0.075*** (0.008)
In co-authorship network			0.349*** (0.012)	0.426*** (0.010)	0.368*** (0.011)	0.421*** (0.010)
In acknowledgment network					0.465*** (0.009)	0.466*** (0.009)
Num.Obs.	16 198	29 050	16 198	29 050	16 198	27 239
R2	0.038	0.037	0.088	0.098	0.222	0.180
R2 Adj.	0.037	0.037	0.087	0.098	0.222	0.180

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table S12: Weighted generalized linear models (frequentist) | DV: h-index

	(1) k2k matching	(1) cem matching	(2) k2k matching	(2) cem matching	(3) k2k matching	(3) cem matching
Intercept	0.962*** (0.023)	0.798*** (0.018)	0.380*** (0.027)	0.127*** (0.021)	0.345*** (0.025)	-0.505*** (0.024)
Male scholar	0.069*** (0.011)	0.133*** (0.008)	0.087*** (0.011)	0.141*** (0.008)	0.068*** (0.009)	0.115*** (0.008)
Senior scholar	0.135*** (0.017)	0.195*** (0.013)	0.151*** (0.017)	0.201*** (0.013)	0.100*** (0.014)	0.145*** (0.013)
Based in North America	0.239*** (0.015)	0.341*** (0.011)	0.262*** (0.015)	0.341*** (0.011)	0.197*** (0.013)	0.269*** (0.011)
Based in Europe	0.251*** (0.014)	0.317*** (0.011)	0.239*** (0.014)	0.297*** (0.011)	0.170*** (0.012)	0.239*** (0.011)
Quantitative scholar	-0.233*** (0.013)	-0.112*** (0.010)	-0.284*** (0.013)	-0.150*** (0.010)	-0.109*** (0.012)	-0.100*** (0.010)
Qualitative scholar	-0.442*** (0.015)	-0.338*** (0.011)	-0.359*** (0.015)	-0.237*** (0.011)	-0.156*** (0.013)	-0.210*** (0.011)
In top 100	0.121*** (0.013)	0.187*** (0.008)	0.133*** (0.013)	0.188*** (0.008)	0.100*** (0.012)	0.143*** (0.008)
In co-authorship network			0.652*** (0.017)	0.753*** (0.012)	0.481*** (0.013)	0.742*** (0.013)
In acknowledgment network					0.618*** (0.011)	0.872*** (0.013)
Num.Obs.	16 198	29 050	16 198	29 050	16 198	27 239
AIC	64 876.1	134 486.4	63 183.3	129 641.2	58 411.2	113 602.8

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table S13: Weighted linear models (frequentist) | DV: Euclid score

	(1) k2k matching	(1) cem matching	(2) k2k matching	(2) cem matching	(3) k2k matching	(3) cem matching
Intercept	2.872*** (0.046)	2.673*** (0.039)	2.144*** (0.050)	1.864*** (0.042)	1.451*** (0.051)	1.339*** (0.044)
Male scholar	-0.021 (0.022)	0.055** (0.017)	0.010 (0.021)	0.069*** (0.017)	0.014 (0.020)	0.042* (0.017)
Senior scholar	0.452*** (0.032)	0.544*** (0.027)	0.479*** (0.031)	0.554*** (0.026)	0.462*** (0.030)	0.486*** (0.026)
Based in North America	0.389*** (0.029)	0.561*** (0.023)	0.423*** (0.028)	0.562*** (0.023)	0.429*** (0.027)	0.475*** (0.023)
Based in Europe	0.368*** (0.026)	0.454*** (0.023)	0.352*** (0.026)	0.423*** (0.022)	0.345*** (0.025)	0.348*** (0.022)
Quantitative scholar	-0.222*** (0.029)	-0.102*** (0.025)	-0.300*** (0.029)	-0.168*** (0.024)	-0.132*** (0.028)	-0.107*** (0.024)
Qualitative scholar	-0.457*** (0.031)	-0.366*** (0.026)	-0.329*** (0.031)	-0.213*** (0.025)	-0.150*** (0.030)	-0.183*** (0.025)
In top 100	0.230*** (0.029)	0.363*** (0.019)	0.248*** (0.028)	0.365*** (0.018)	0.247*** (0.026)	0.303*** (0.018)
In co-authorship network			0.843*** (0.027)	0.951*** (0.021)	0.878*** (0.025)	0.938*** (0.021)
In acknowledgment network					0.817*** (0.021)	0.829*** (0.020)
Num.Obs.	16 191	29 042	16 191	29 042	16 191	27 231
R2	0.044	0.060	0.100	0.122	0.180	0.176
R2 Adj.	0.043	0.060	0.100	0.122	0.179	0.176

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table S14: Weighted generalized linear models (frequentist) | DV: Euclid score

	(1) k2k matching	(1) cem matching	(2) k2k matching	(2) cem matching	(3) k2k matching	(3) cem matching
Intercept	2.872*** (0.046)	2.673*** (0.039)	2.144*** (0.050)	1.864*** (0.042)	1.451*** (0.051)	1.339*** (0.044)
Male scholar	-0.021 (0.022)	0.055** (0.017)	0.010 (0.021)	0.069*** (0.017)	0.014 (0.020)	0.042* (0.017)
Senior scholar	0.452*** (0.032)	0.544*** (0.027)	0.479*** (0.031)	0.554*** (0.026)	0.462*** (0.030)	0.486*** (0.026)
Based in North America	0.389*** (0.029)	0.561*** (0.023)	0.423*** (0.028)	0.562*** (0.023)	0.429*** (0.027)	0.475*** (0.023)
Based in Europe	0.368*** (0.026)	0.454*** (0.023)	0.352*** (0.026)	0.423*** (0.022)	0.345*** (0.025)	0.348*** (0.022)
Quantitative scholar	-0.222*** (0.029)	-0.102*** (0.025)	-0.300*** (0.029)	-0.168*** (0.024)	-0.132*** (0.028)	-0.107*** (0.024)
Qualitative scholar	-0.457*** (0.031)	-0.366*** (0.026)	-0.329*** (0.031)	-0.213*** (0.025)	-0.150*** (0.030)	-0.183*** (0.025)
In top 100	0.230*** (0.029)	0.363*** (0.019)	0.248*** (0.028)	0.365*** (0.018)	0.247*** (0.026)	0.303*** (0.018)
In co-authorship network			0.843*** (0.027)	0.951*** (0.021)	0.878*** (0.025)	0.938*** (0.021)
In acknowledgment network					0.817*** (0.021)	0.829*** (0.020)
Num.Obs.	16 191	29 042	16 191	29 042	16 191	27 231
AIC	56 272.6	101 692.7	55 209.1	99 658.9	54 066.3	92 212.9

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table S15: Weighted linear models (Bayesian) | DV: h-index

	(1) k2k matching	(1) cem matching	(2) k2k matching	(2) cem matching	(3) k2k matching	(3) cem matching
Intercept	1.062*** (0.019)	0.949*** (0.016)	0.764*** (0.021)	0.597*** (0.018)	0.484*** (0.020)	0.425*** (0.018)
Male scholar	0.040*** (0.009)	0.087*** (0.008)	0.052*** (0.009)	0.092*** (0.007)	0.043*** (0.008)	0.062*** (0.007)
Senior scholar	0.028* (0.014)	0.068*** (0.012)	0.039** (0.013)	0.074*** (0.012)	0.018 (0.012)	0.020+ (0.012)
Based in North America	0.134*** (0.012)	0.238*** (0.010)	0.148*** (0.012)	0.236*** (0.010)	0.116*** (0.011)	0.153*** (0.010)
Based in Europe	0.140*** (0.011)	0.196*** (0.010)	0.133*** (0.011)	0.180*** (0.010)	0.114*** (0.010)	0.125*** (0.010)
Quantitative scholar	-0.057*** (0.012)	0.006 (0.010)	-0.088*** (0.012)	-0.037*** (0.010)	-0.084*** (0.011)	-0.096*** (0.010)
Qualitative scholar	-0.118*** (0.013)	-0.079*** (0.011)	-0.066*** (0.013)	-0.023* (0.011)	-0.066*** (0.012)	-0.108*** (0.010)
In top 100	0.030* (0.012)	0.111*** (0.009)	0.038** (0.012)	0.110*** (0.008)	0.039*** (0.011)	0.075*** (0.008)
In co-authorship network			0.346*** (0.011)	0.432*** (0.009)	0.359*** (0.010)	0.413*** (0.009)
In acknowledgment network					0.484*** (0.008)	0.490*** (0.009)
Num.Obs.	16 826	29 946	16 826	29 946	16 826	28 039
R2	0.017	0.036	0.068	0.099	0.223	0.192
R2 Adj.	0.017	0.036	0.068	0.099	0.222	0.192

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table S16: Weighted generalized linear models (Bayesian) | DV: h-index

	(1) k2k matching	(1) cem matching	(2) k2k matching	(2) cem matching	(3) k2k matching	(3) cem matching
Intercept	0.737*** (0.022)	0.571*** (0.017)	0.152*** (0.027)	-0.094*** (0.021)	0.311*** (0.024)	-0.560*** (0.023)
Male scholar	0.088*** (0.011)	0.155*** (0.008)	0.106*** (0.011)	0.162*** (0.008)	0.068*** (0.009)	0.118*** (0.008)
Senior scholar	0.148*** (0.017)	0.213*** (0.013)	0.165*** (0.017)	0.219*** (0.013)	0.096*** (0.014)	0.146*** (0.013)
Based in North America	0.259*** (0.015)	0.403*** (0.011)	0.283*** (0.015)	0.399*** (0.011)	0.168*** (0.013)	0.274*** (0.011)
Based in Europe	0.267*** (0.014)	0.344*** (0.011)	0.255*** (0.014)	0.322*** (0.011)	0.164*** (0.011)	0.238*** (0.011)
Quantitative scholar	-0.042** (0.013)	0.054*** (0.010)	-0.092*** (0.013)	-0.002 (0.010)	-0.063*** (0.011)	-0.072*** (0.010)
Qualitative scholar	-0.220*** (0.015)	-0.164*** (0.011)	-0.135*** (0.015)	-0.079*** (0.011)	-0.100*** (0.013)	-0.185*** (0.011)
In top 100	0.084*** (0.014)	0.193*** (0.008)	0.097*** (0.014)	0.192*** (0.008)	0.073*** (0.012)	0.143*** (0.008)
In co-authorship network			0.657*** (0.017)	0.769*** (0.012)	0.472*** (0.013)	0.737*** (0.013)
In acknowledgment network					0.629*** (0.011)	0.903*** (0.012)
Num.Obs.	16 826	29 946	16 826	29 946	16 826	28 039
AIC	64 396.7	135 435.8	62 583.0	130 740.8	61 426.0	119 115.1

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table S17: Weighted linear models (Bayesian) | DV: Euclid score

	(1) k2k matching	(1) cem matching	(2) k2k matching	(2) cem matching	(3) k2k matching	(3) cem matching
Intercept	2.488*** (0.043)	2.277*** (0.036)	1.777*** (0.048)	1.488*** (0.039)	1.258*** (0.046)	1.168*** (0.039)
Male scholar	-0.011 (0.022)	0.087*** (0.017)	0.018 (0.021)	0.099*** (0.016)	0.002 (0.020)	0.046** (0.016)
Senior scholar	0.479*** (0.032)	0.553*** (0.026)	0.507*** (0.031)	0.565*** (0.026)	0.468*** (0.029)	0.473*** (0.026)
Based in North America	0.454*** (0.029)	0.653*** (0.023)	0.490*** (0.028)	0.648*** (0.022)	0.430*** (0.026)	0.490*** (0.022)
Based in Europe	0.405*** (0.026)	0.487*** (0.022)	0.387*** (0.025)	0.452*** (0.021)	0.352*** (0.024)	0.344*** (0.021)
Quantitative scholar	0.108*** (0.027)	0.222*** (0.023)	0.033 (0.026)	0.127*** (0.022)	0.040 (0.025)	0.014 (0.022)
Qualitative scholar	-0.120*** (0.029)	-0.032 (0.024)	0.005 (0.029)	0.094*** (0.023)	0.004 (0.027)	-0.069** (0.023)
In top 100	0.210*** (0.028)	0.373*** (0.019)	0.230*** (0.028)	0.372*** (0.018)	0.231*** (0.026)	0.302*** (0.018)
In co-authorship network			0.825*** (0.026)	0.966*** (0.021)	0.849*** (0.025)	0.928*** (0.021)
In acknowledgment network					0.898*** (0.020)	0.911*** (0.019)
Num.Obs.	16 821	29 937	16 821	29 937	16 821	28 030
R2	0.041	0.068	0.094	0.132	0.193	0.197
R2 Adj.	0.040	0.068	0.093	0.132	0.192	0.196

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table S18: Weighted generalized linear models (Bayesian) | DV: Euclid score

	(1) k2k matching	(1) cem matching	(2) k2k matching	(2) cem matching	(3) k2k matching	(3) cem matching
Intercept	2.488*** (0.043)	2.277*** (0.036)	1.777*** (0.048)	1.488*** (0.039)	1.258*** (0.046)	1.168*** (0.039)
Male scholar	-0.011 (0.022)	0.087*** (0.017)	0.018 (0.021)	0.099*** (0.016)	0.002 (0.020)	0.046** (0.016)
Senior scholar	0.479*** (0.032)	0.553*** (0.026)	0.507*** (0.031)	0.565*** (0.026)	0.468*** (0.029)	0.473*** (0.026)
Based in North America	0.454*** (0.029)	0.653*** (0.023)	0.490*** (0.028)	0.648*** (0.022)	0.430*** (0.026)	0.490*** (0.022)
Based in Europe	0.405*** (0.026)	0.487*** (0.022)	0.387*** (0.025)	0.452*** (0.021)	0.352*** (0.024)	0.344*** (0.021)
Quantitative scholar	0.108*** (0.027)	0.222*** (0.023)	0.033 (0.026)	0.127*** (0.022)	0.040 (0.025)	0.014 (0.022)
Qualitative scholar	-0.120*** (0.029)	-0.032 (0.024)	0.005 (0.029)	0.094*** (0.023)	0.004 (0.027)	-0.069** (0.023)
In top 100	0.210*** (0.028)	0.373*** (0.019)	0.230*** (0.028)	0.372*** (0.018)	0.231*** (0.026)	0.302*** (0.018)
In co-authorship network			0.825*** (0.026)	0.966*** (0.021)	0.849*** (0.025)	0.928*** (0.021)
In acknowledgment network					0.898*** (0.020)	0.911*** (0.019)
Num.Obs.	16 821	29 937	16 821	29 937	16 821	28 030
AIC	166 234.1	309 998.4	165 173.7	307 822.2	163 709.7	288 272.7

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

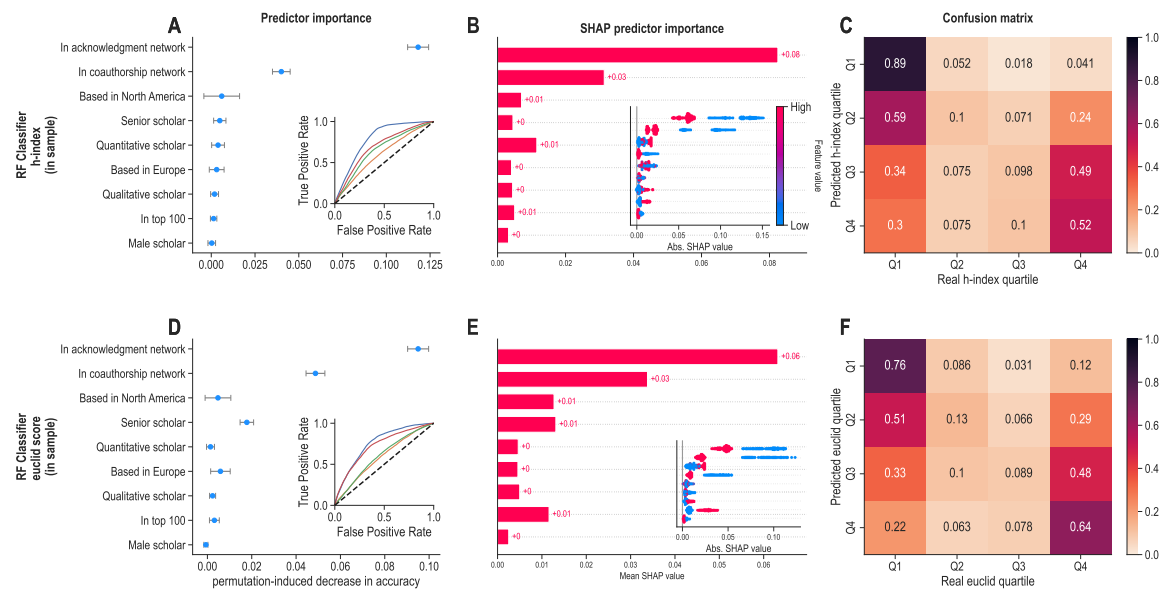


Fig. S35. Random forest classifier for h-index and Euclid scores quartile prediction. This figure is an extension to Fig. 3 in the main text. Upper row for h-index. Lower row for Euclid scores. (A-D) show the permutation error increase importance for the variables used in the model (insets display the ROC-AUC for each quartile). (B-E) show the mean SHAP values, which quantify the contribution of each variable to the prediction made by the model (full distribution in insets; color encodes high to low values in each variable, the further from 0, the higher the importance of each value). (C-F) shows the confusion matrix between predicted and observed values, the darker the color the higher the accuracy.

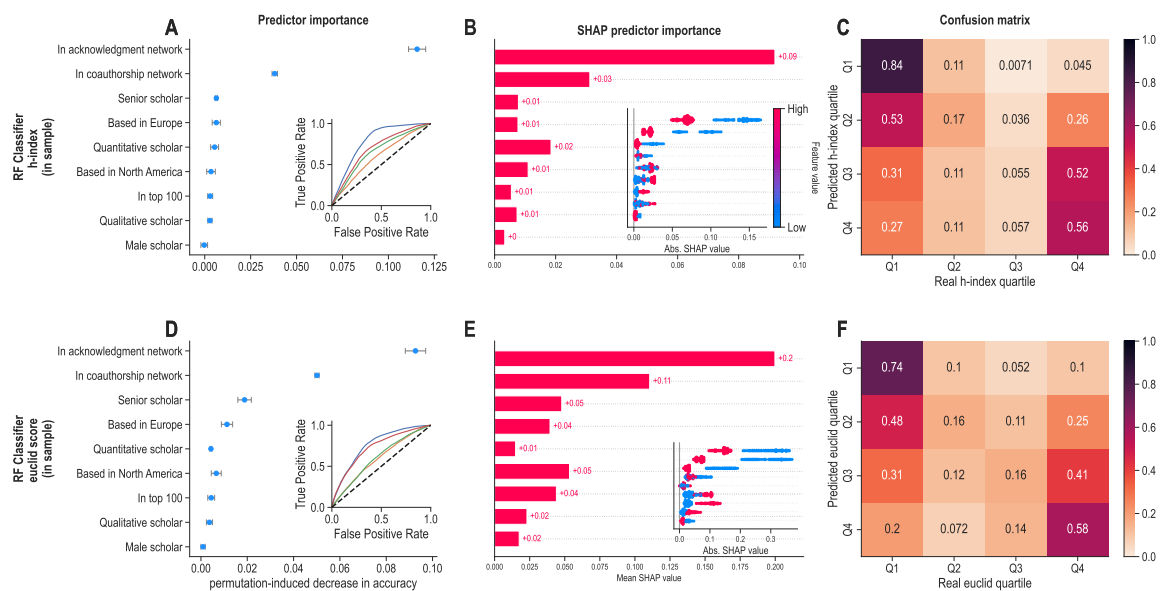


Fig. S36. XGBoost classifier for h-index and Euclid scores quartile prediction. This figure is an extension to Fig. 3 in the main text. Upper row for h-index. Lower row for Euclid scores. (A-D) show the permutation error increase importance for the variables used in the model (insets display the ROC-AUC for each quartile). (B-E) show the mean SHAP values, which quantify the contribution of each variable to the prediction made by the model (full distribution in insets; color encodes high to low values in each variable, the further from 0, the higher the importance of each value). (C-F) shows the confusion matrix between predicted and observed values, the darker the color the higher the accuracy.

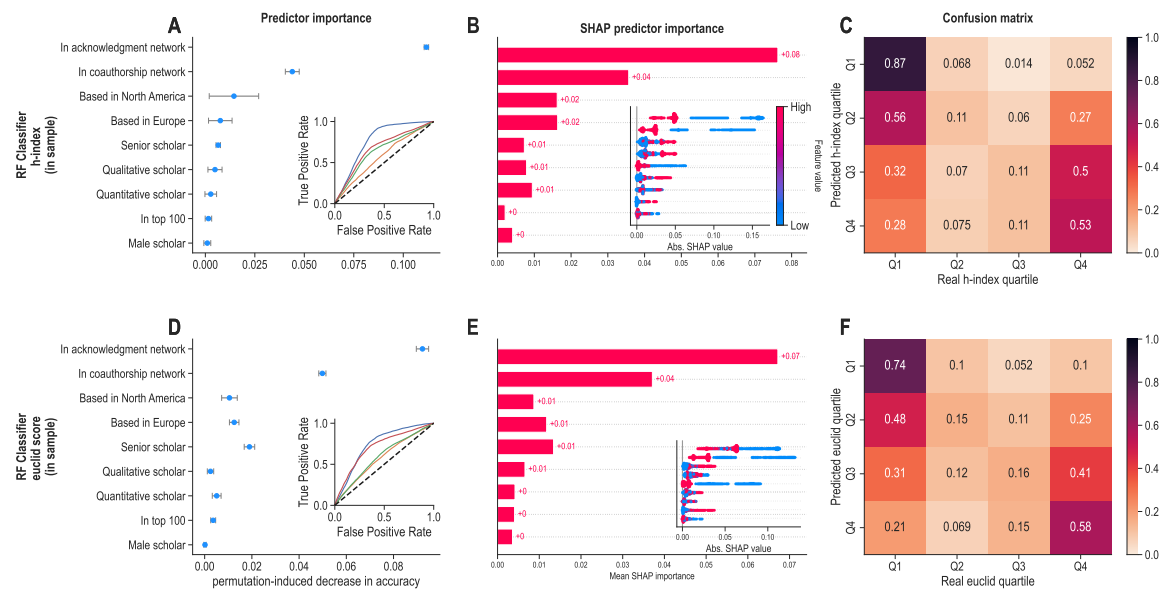


Fig. S37. Support Vector Classifier for h-index and Euclid scores quartile prediction. This figure is an extension to Fig. 3 in the main text. Upper row for h-index. Lower row for Euclid scores. (A-D) show the permutation error increase importance for the variables used in the model (insets display the ROC-AUC for each quartile). (B-E) show the mean SHAP values, which quantify the contribution of each variable to the prediction made by the model (full distribution in insets; color encodes high to low values in each variable, the further from 0, the higher the importance of each value). (C-F) shows the confusion matrix between predicted and observed values, the darker the color the higher the accuracy.

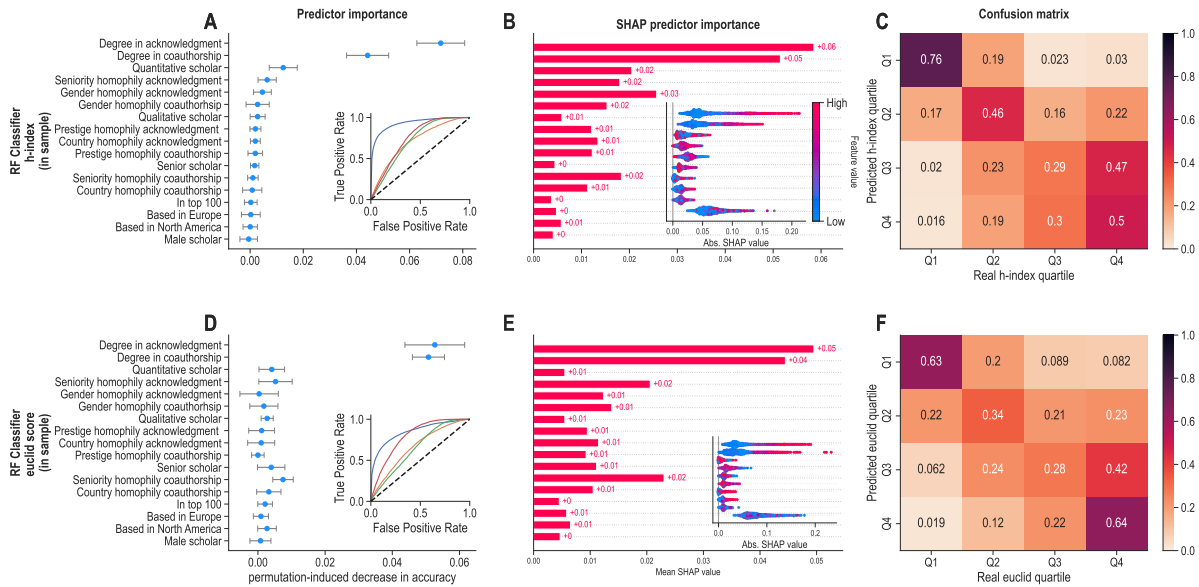


Fig. S38. Random forest classifier for h-index and Euclid scores quartile prediction. This figure is an extension to Fig. 4 in the main text. Upper row for h-index. Lower row for Euclid scores. (A-D) show the permutation error increase importance for the variables used in the model (insets display the ROC-AUC for each quartile). (B-E) show the mean SHAP values, which quantify the contribution of each variable to the prediction made by the model (full distribution in insets; color encodes high to low values in each variable, the further from 0, the higher the importance of each value). (C-F) shows the confusion matrix between predicted and observed values, the darker the color the higher the accuracy.

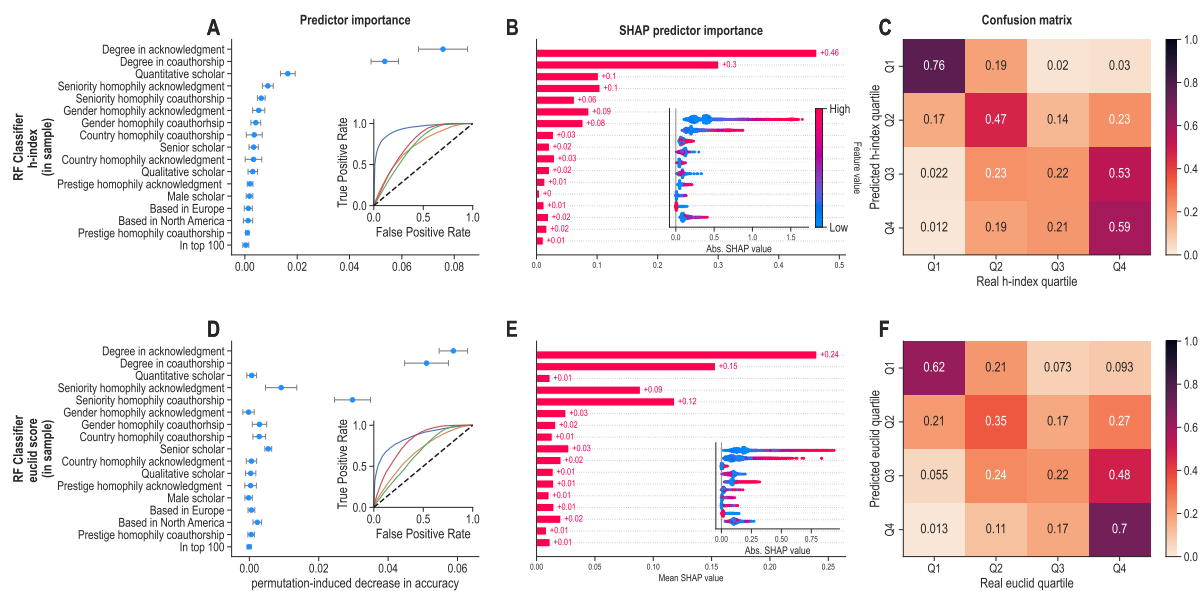


Fig. S39. XGBoost classifier for h-index and Euclid scores quartile prediction. This figure is an extension to Fig. 4 in the main text. Upper row for h-index. Lower row for Euclid scores. (A-D) show the permutation error increase importance for the variables used in the model (insets display the ROC-AUC for each quartile). (B-E) show the mean SHAP values, which quantify the contribution of each variable to the prediction made by the model (full distribution in insets; color encodes high to low values in each variable, the further from 0, the higher the importance of each value). (C-F) shows the confusion matrix between predicted and observed values, the darker the color the higher the accuracy.

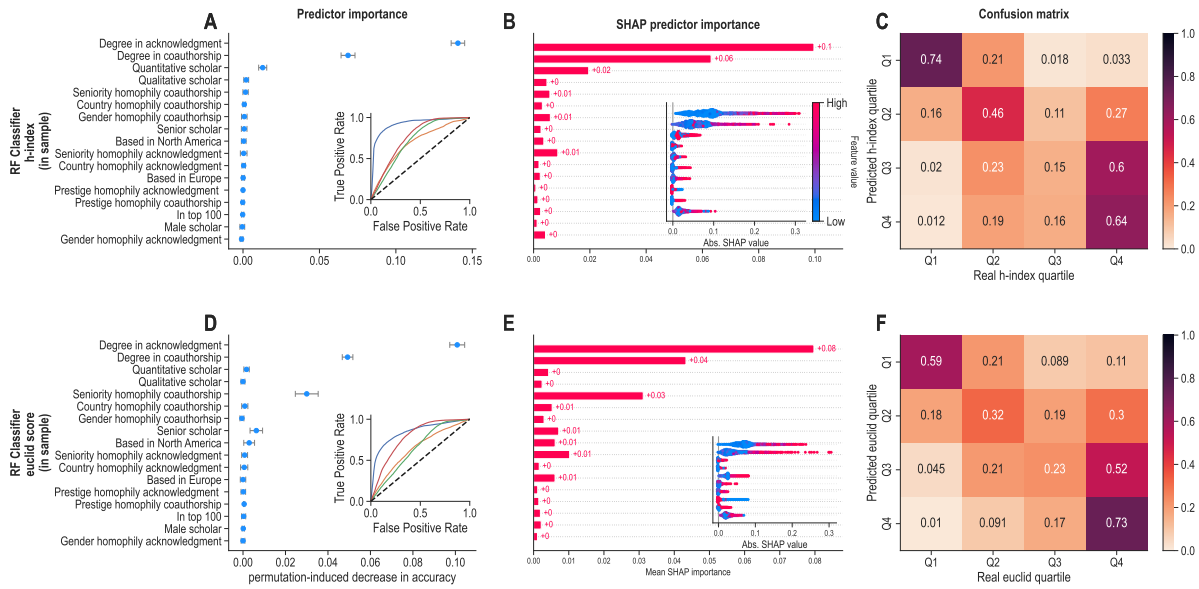


Fig. S40. Support Vector Classifier for h-index and Euclid scores quartile prediction. This figure is an extension to Fig. 4 in the main text. Upper row for h-index. Lower row for Euclid scores. (A-D) show the permutation error increase importance for the variables used in the model (insets display the ROC-AUC for each quartile). (B-E) show the mean SHAP values, which quantify the contribution of each variable to the prediction made by the model (full distribution in insets; color encodes high to low values in each variable, the further from 0, the higher the importance of each value). (C-F) shows the confusion matrix between predicted and observed values, the darker the color the higher the accuracy.

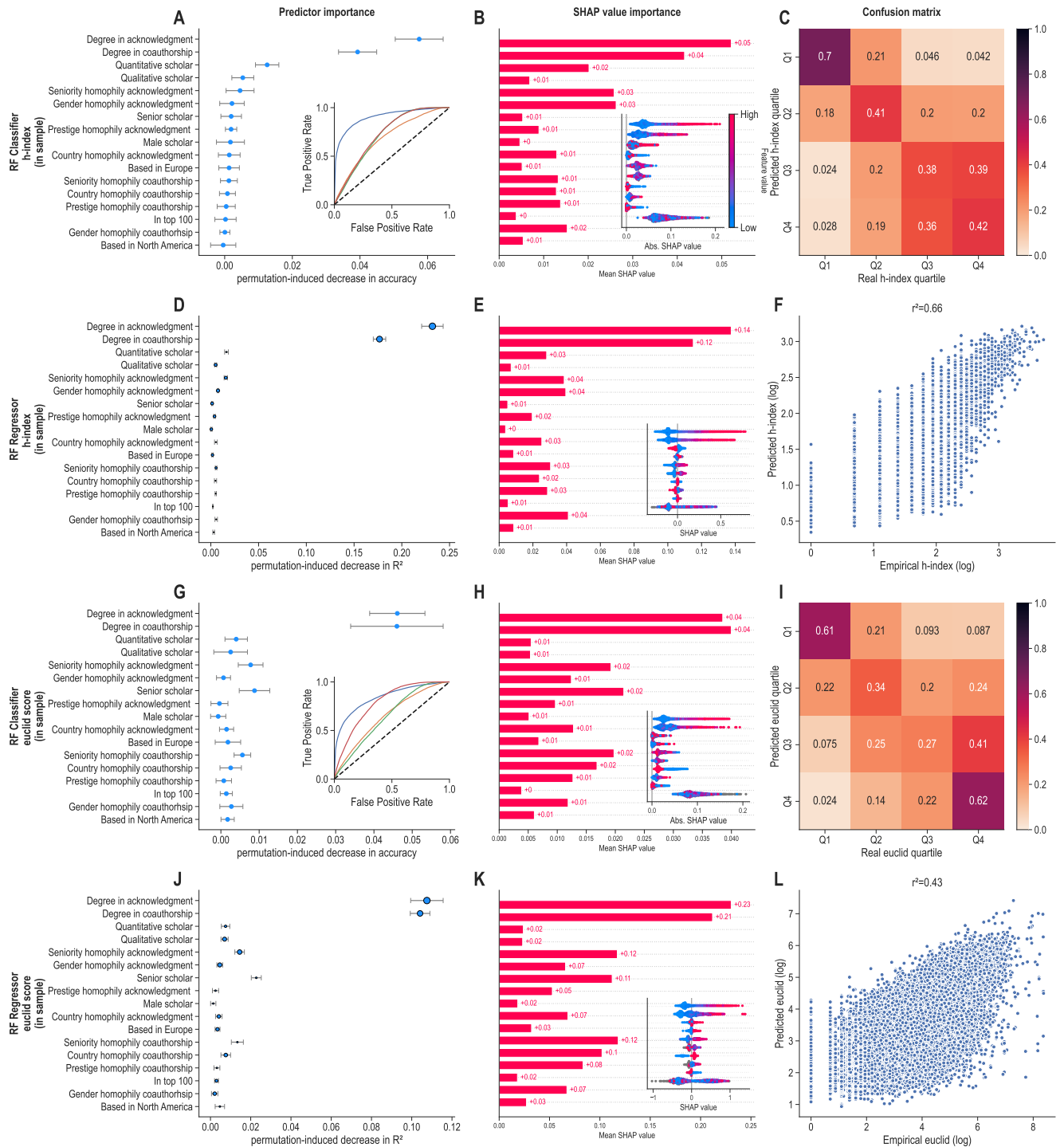


Fig. S41. Random forest regression and quartile classifier for h-index and Euclid scores with a 5-publications threshold. This figure is an extension to Fig. 4 in the main text and Fig. S38 using a 5-paper threshold to filter scholars (see also section 8). (A, D, G, J) show the permutation error increase importance for the variables used in the regression and classifier models (insets show the ROC-AUC curves associated to each quartile prediction). (B, E, H, K) show the mean SHAP values, which quantify the contribution of each variable to the prediction made by the model (full distribution in insets; color encodes high to low values in each variable, the further from 0, the higher the importance of each value). (C, F, I, L) show the confusion matrix or correlation between predicted and observed values (the darker the color in the matrices or the higher the correlation coefficient, the higher the accuracy).

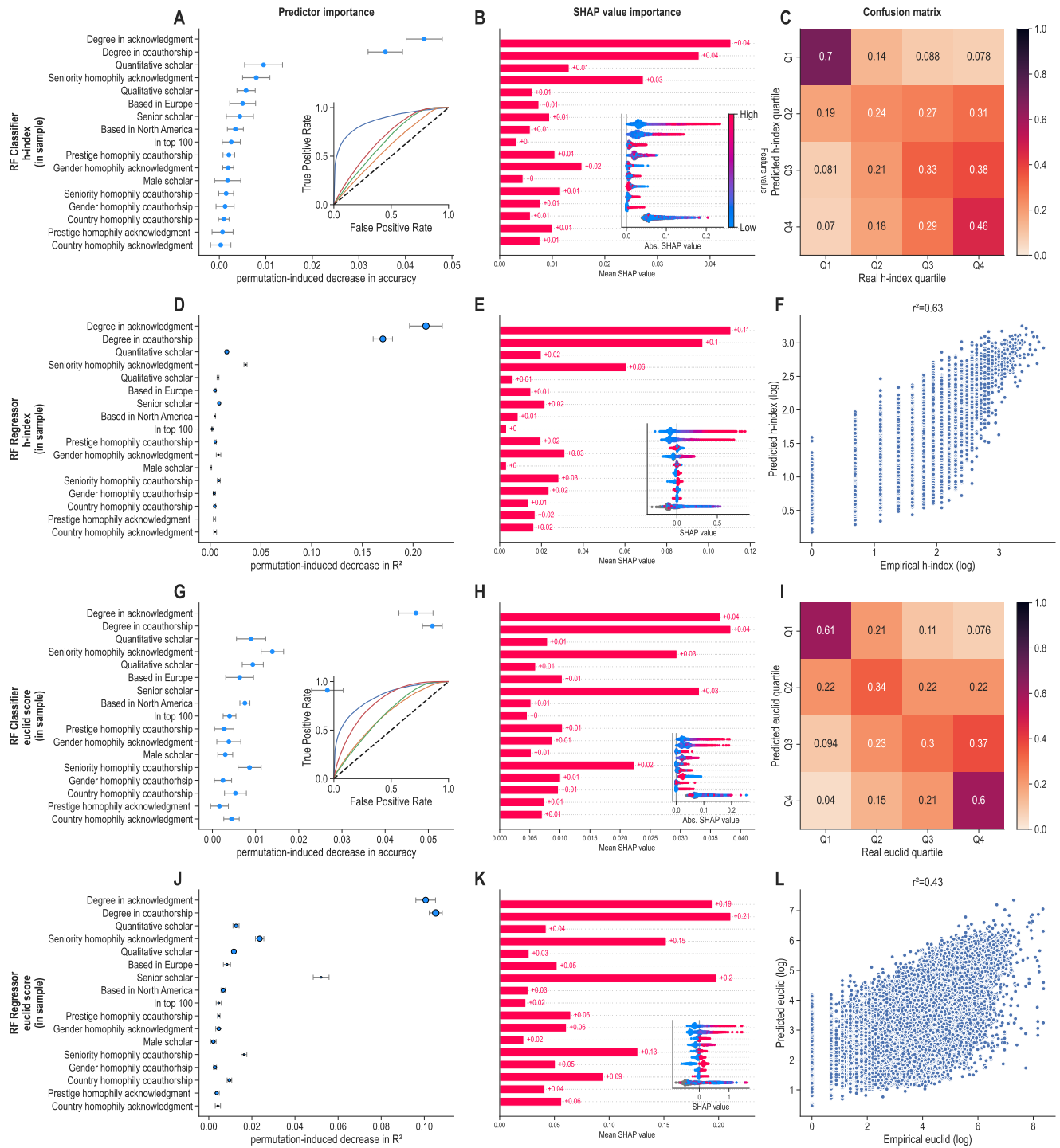


Fig. S42. Random forest regression and quartile classifier for h-index and Euclid scores without publication threshold. This figure is an extension to Fig. 4 in the main text and Figs. S38 and S41 using no threshold to filter scholars with low publications (see also section 8). (A, D, G, J) show the permutation error increase importance for the variables used in the regression and classifier models (insets show the ROC-AUC curves associated to each quartile prediction). (B, E, H, K) show the mean SHAP values, which quantify the contribution of each variable to the prediction made by the model (full distribution in insets; color encodes high to low values in each variable, the further from 0, the higher the importance of each value). (C, F, I, L) show the confusion matrix or correlation between predicted and observed values (the darker the color in the matrices or the higher the correlation coefficient, the higher the accuracy).

References

1. DL Teele, K Thelen, Gender in the journals: Publication patterns in political science. *Polit. Sci. & Polit.* **50**, 433–447 (2017).
2. M Newman, *Networks. An Introduction.* (Oxford University Press, Oxford), (2018).
3. S Wasserman, K Faust, *Social Network Analysis: Methods and Applications.* (Cambridge University Press, Cambridge), (1994).
4. M McPherson, L Smith-Lovin, JM Cook, Birds of a feather: Homophily in social networks. *Annu. Rev. Sociol.* **27**, 415–444 (2001).
5. YS Wang, CJ Lee, JD West, CT Bergstrom, EA Erosheva, Gender-based homophily in collaborations across a heterogeneous scholarly landscape. *PLOS ONE* **18**, e0283106 (2023).
6. M Torre, JA Prieto-Alonso, I Ucar, The uneven effects of gender parity: Trends in gender homophily in scientific publications, 1980–2019. *Soc. Sci. Res.* **132**, 103228 (2025).
7. MEJ Newman, Assortative mixing in networks. *Phys. Rev. Lett.* **89**, 208701 (2002).
8. J Huang, AJ Gates, R Sinatra, AL Barabási, Historical comparison of gender inequality in scientific careers across countries and disciplines. *Proc. Natl. Acad. Sci.* **117**, 4609–4616 (2020).
9. L Danús, C Muntaner, A Krauss, M Sales-Pardo, R Guimerà, Differences in collaboration structures and impact among prominent researchers in europe and north america. *EPJ Data Sci.* **12**, 12 (2023).
10. V Larivière, Y Gingras, On the relationship between interdisciplinarity and scientific impact. *J. Am. Soc. for Inf. Sci. Technol.* **61**, 126–131 (2010).
11. L Breiman, Random forests. *Mach. Learn.* **45**, 5–32 (2001).
12. T Chen, C Guestrin, Xgboost: A scalable tree boosting system (2016).
13. C Cortes, V Vapnik, Support-vector networks. *Mach. Learn.* **20**, 273–297 (1995).
14. SM Lundberg, SI Lee, A unified approach to interpreting model predictions in *Advances in Neural Information Processing Systems*, eds. I Guyon, et al. (Curran Associates, Inc.), Vol. 30, (2017).